NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

MIL-PRF-39016E
18 July 1994
SUPERSEDING
MIL-R-39016D
19 September 1986

PERFORMANCE SPECIFICATION

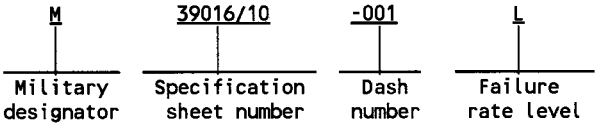
RELAYS, ELECTROMAGNETIC, ESTABLISHED RELIABILITY,
GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for electromagnetic, hermetically sealed relays for use in electronic and communication-type equipment (see 6.1). These relays are designed to operate over the full range from low level to power switching with contact ratings up to 5 amperes alternating current (ac) or direct current (dc). The failure rate level is established at a confidence level of 90 percent for qualification and 60 percent for maintenance of qualification based on 100,000 cycles at +125°C under the rated load conditions specified herein. CAUTION: The use of any coil voltage less than the rated coil voltage will compromise the operation of the relay. CAUTION: Pick-up, hold, and dropout voltages are for test purposes only and are not to be used as design criteria. For additional application and caution information, see 6.1.

1.2 Part or Identifying Number (PIN). The PIN shall consist of the letter "M", the basic number of the specification sheet, an assigned dash number (see 3.1), and a suffix letter designating failure rate level (see table I) as shown in the following example:



1.3 Failure rate (FR) level designation. The failure rate level designation is shown in table I (see 4.6).

TABLE I. Failure rate level designation.

Failure rate level designation	Failure rate level (percent per 10,000 cycles)
L	3.0
M	1.0
P	0.1
R	0.01

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Electronics Supply Center, ATTN: DESC-ELDM, 1507 Wilmington Pike, Dayton, OH 45444-5765, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

J-W-1177 - Wire, Magnet, Electrical.
QQ-N-290 - Nickel Plating (Electrodeposited).
ZZ-R-765 - Rubber, Silicone: Low- and High-Temperature and Tear Resistant.

MILITARY

MIL-I-10 - Insulating Compound, Electrical, Ceramic, Class L.
MIL-M-14 - Molding Plastics and Molded Plastic Parts, Thermosetting.
MIL-P-116 - Preservation, Methods of.
MIL-P-997 - Plastic Material, Laminated, Thermosetting, Electrical Insulation: Sheets, Glass Cloth.
MIL-P-15037 - Plastic Sheet, Laminated, Thermosetting, Glass-Cloth, Melamine-Resin.
MIL-P-15047 - Plastic-Material, Laminated Thermosetting, Sheets, Nylon Fabric Base, Phenolic-Resin.
MIL-G-45204 - Gold Plating, Electrodeposited.

STANDARDS

FEDERAL

FED-STD-123 - Marking for Shipment.

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454 - Standard General Requirements for Electronic Equipment.
MIL-STD-690 - Failure Rate Sampling Plans and Procedures.
MIL-STD-750 - Test Methods for Semiconductor Devices.
MIL-STD-790 - Product Assurance Program for Electronic Parts Specifications.
MIL-STD-883 - Test Methods and Procedures for Microelectronics.
MIL-STD-1276 - Leads for Electronic Component Parts.
MIL-STD-1285 - Marking of Electrical And Electronic Parts.
MIL-STD-1346 - Relays Selection and Application
MIL-STD-1686 - Electrostatic Discharge Control Program for Protection of Electrical and Electronic Parts.
MIL-STD-2073-1 - DoD Materiel Procedures for the Development and Application of Packaging Requirements.

HANDBOOKS

MIL-HDBK-263 - Electrostatic Discharge Control Handbook for Protection of Electrical and Electronic Parts, Assemblies and Equipment.
MIL-HDBK-338 - Electronic Reliability Design Handbook.

(See supplement 1 for list of associated specifications.)

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Defense Printing Service Detachment Office, Building 4D (Customer Service), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

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2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI Y32.2 - Graphic Symbols for Electric and Electronic Diagrams.

(Application for copies should be addressed to the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.)

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

EIA-557 - Statistical Process Control Systems.

(Application for copies should be addressed to the Electronic Industries Association, 2001 Pennsylvania Avenue, NW, Washington, DC 20006.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern.

3.2 Qualification. Relays furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list (QPL) at the time of award of contract (see 4.4 and 6.3). Authorized distributors which are approved to MIL-STD-790 distributor requirements by the QPL manufacturers are listed in the QPL.

3.3 Reliability and quality.

3.3.1 Reliability. Reliability of relays furnished under this specification shall be established and maintained in accordance with the requirements and procedures specified in MIL-STD-790 and MIL-STD-690 with details and exceptions specified in 4.1.2, 4.4.4, and 4.5.

3.3.2 Quality.

3.3.2.1 Statistical process control (SPC). The contractor shall implement and use statistical process control techniques in the manufacturing process for parts covered by this specification. The SPC program shall be developed and maintained in accordance with EIA-557. The SPC program shall be documented and maintained as part of the overall product assurance program as specified in MIL-STD-790. The implementation date for statistical process control shall be 12 months from the date of revision E of this specification.

3.4 Materials. Materials shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the relays to meet the performance requirements of this specification. Materials used shall be self-extinguishing, and shall not support combustion, give off noxious gases in harmful quantities, give off gases in quantities sufficient to cause explosion of sealed enclosures, cause contamination of the contacts or other parts of the relay, or form current-carrying tracks when subjected to any of the tests specified herein. The selection of materials shall be such as to provide maximum shelf life. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.4.1 Plastic. Laminated plastic material shall conform to MIL-P-997, MIL-P-15037, or MIL-P-15047. Molded plastic material shall conform to MIL-M-14 Cotton-filled or wood-flour-filled materials shall not be used.

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3.4.2 Ceramic. Ceramic insulating material shall conform to MIL-I-10, grade L422 or higher. Ceramic used for external surfaces shall be glazed.

3.4.3 Fungus-resistant. Materials used in the construction of relays shall be fungus inert (see requirement 4 of MIL-STD-454).

3.4.4 Metals. Metals shall be of a corrosion-resistant type or shall be plated or treated to resist corrosion.

3.4.4.1 Plated finishes.

a. Use of tin plating is prohibited internally and externally (see 6.6.4.1). Use of tin-lead finishes are acceptable provided that the minimum lead content is 3 percent.

b. Use of zinc plating is prohibited internally and externally.

c. Use of cadmium plating is prohibited internally and externally.

3.4.4.2 Dissimilar metals. When dissimilar metals are used in intimate contact with each other, protection against electrolysis and corrosion shall be provided. The use of dissimilar metals in contact, which tends toward active electrolytic corrosion (particularly brass, copper, or steel used in contact with aluminum or aluminum alloy), is not acceptable. However, metal spraying or metal plating of dissimilar base metals to provide similar or suitable abutting surfaces is permitted. Dissimilar metals should be as defined in 6.6 through 6.6.4, inclusive. In hermetic seals, the 0.25 volt difference between the header material and the housing material is not applicable.

3.4.5 Magnet wire. Magnet wire shall conform to J-W-1177. AWG wire sizes used in relays but not listed in J-W-1177 shall meet the basic material and coating requirements of J-W-1177 as a minimum.

3.4.6 Rubber. Rubber shall conform to ZZ-R-765.

3.4.7 Mercury. The use of mercury or mercury compounds is prohibited.

3.4.8 Magnesium. The use of magnesium or magnesium alloys is prohibited (not applicable to contacts).

3.4.9 Silicone or silicone compounds. The use of silicone (see 6.8) or silicone compounds for any purpose is prohibited.

3.5 Design and construction. Relays shall be of the design, construction, weight, and physical dimensions specified (see 3.1).

3.5.1 Case. Unless otherwise specified (see 3.1), the case shall not be electrically connected to the contacts or coil; however, it may be used as part of the magnetic circuit.

3.5.1.1 Case grounding When specified (see 3.1), means for connecting the relay case to ground shall be provided.

3.5.2 Sealing process. Relays shall be dried, degassed, and backfilled with an atmosphere and sealed by welding such that the requirements of this specification are met. Adjunct sealant (see 6.8), if used, must comply with the following characteristics:

a. Shall not extend above 20 percent of the length of the exposed terminals above the glass meniscus.

b. Trace color is permitted if it is a natural result of the sealant process.

c. Shall form, after curing, a permanent nonconductive, noncracking seal under all relay environments.

3.5.3 Contacts. Contacts shall have load ratings and arrangements (see MIL-STD-1285) as specified (see 3.1) and unless otherwise specified (see 3 1), shall be capable of carrying the maximum rated current continuously as well as making and breaking the specified current under all environmental conditions specified herein.

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3.5.4 Coils. Coils shall be adequately insulated electrically from the contacts and the case. The resistance and rated voltage (or current) shall be as specified (see 3.1). Coils shall be designed for continuous operation at maximum rated voltage and temperature, unless otherwise specified (see 3.1).

3.5.4.1 Terminal identification. When specified (see 3.1), a bead of contrasting color shall be used to designate the X1 (positive, if applicable) terminal (see figure 1 and MIL-STD-1285).

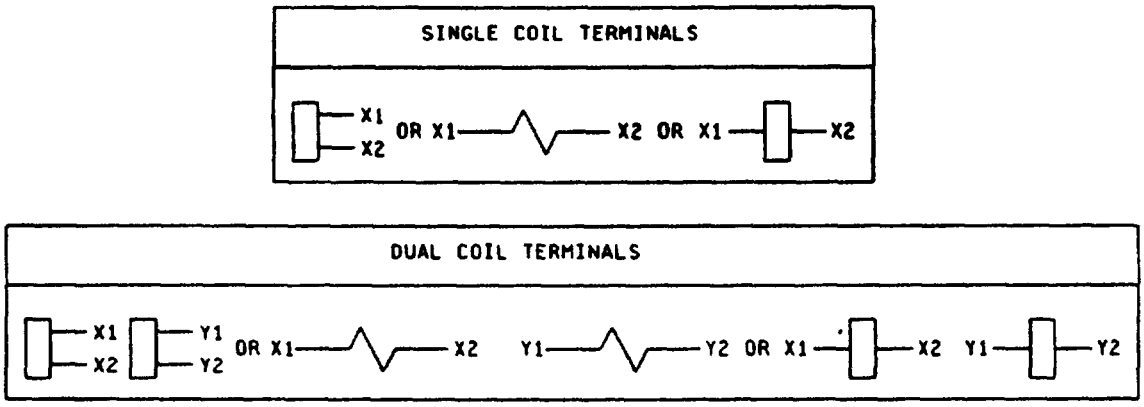


FIGURE 1. Symbols and marking for terminals.

3.5.4.2 Latching relays. Latching relays with two coils shall be so designed that if both coils are energized simultaneously, the contacts should not achieve a neutral position (both the normally closed and normally open contacts are open). The relay shall be screened as specified in 3.12.7 and 4.8.8.7. Specified dropout value (voltage or current) and release time are not applicable to latching relays (see 6.1).

3.5.5 Circuit diagram. The circuit diagram as specified (see 3.1), shall be a terminal view. Circuit symbols shall be in accordance with ANSI Y32.2. For relays without an orientation tab, the circuit diagram, as specified (see 3.1), shall be oriented so that when the relay is held with the circuit diagram right side up as shown (see 3.1), and rotated away from the viewer about a horizontal axis through the diagram until the header terminals face the viewer, then each terminal shall be in the location shown in the circuit diagram.

3.5.6 Mounting means (see 3.1).

3.5.6.1 Bracket. Mounting brackets shall be an integral part of the relay, securely attached thereto in a manner to prevent any movement between the relay and the mounting bracket.

3.5.7 Terminals (see 3.1). Terminals shall be as specified herein. Manufacturer may supply hot solder dipped terminals provided that the hot solder dipping process has been approved by the qualifying activity and when specified on the individual purchase order (see 6.2.1a). Solder dipped terminals may be .002 inch (0.050 mm) larger than the maximum dimension specified (see 3.1).

3.5.7.1 Solder-lug terminals. Solder-lug terminals shall be designed to accommodate two conductors, each rated to carry the maximum rated current of the contact or coil terminated.

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3.5.7.2 Wire leads. Wire leads shall be as specified (see 3.1). Optional, shortened wire leads may be supplied when specified on the individual purchase order (see 3.1 and 6.2.1b).

3.5.7.2.1 Wire leads, solder pin (SP). Solder pin wire leads shall be as specified (see 3.1).

3.5.7.3 Plug-in termination. Plug-in terminations shall conform to the arrangements or dimensions as specified (see 3.1). The mounting arrangement of the relay shall be so designed that the entire weight of the relay will be suspended and the stability of its mounting will be provided by an auxiliary mounting means other than the electrical terminals of a socket (see 3.1). Plug-in terminals shall be gold plated in accordance with MIL-G-45204, type II, class I, with a nickel underplating that shall be in accordance with QQ-N-290 and 50 to 150 microinches thick.

3.5.7.4 Solder dip (retinning) leads. The manufacturer may solder dip/retin the leads of product supplied to this specification provided the solder dip process has been approved by the qualifying activity.

3.5.7.4.1 Qualifying activity approval. Approval of the solder dip process will be based on one of the following options (Note: Solder dip of gold-plated plug-in leads is not allowed.) All visual examination criteria shall be in accordance with method 208 of MIL-STD-202:

a. When the original lead finish qualified was hot solder dip lead finish 52 of MIL-STD-1276 (The 200-microinch thickness is not applicable). The manufacturer shall use the same solder dip process for retinning as is used in the original manufacture of the product.

b. When the lead originally qualified was not hot solder dip lead finish 52 as prescribed above, approval for the process to be used for solder dip shall be based on the following test procedure:

(1) Six samples for each style and lead finish are subjected to the manufacturer's solder dip process. Following the solder dip process, the relays shall be subjected to groups A2 and A4 inspections.

(2) Three of the six samples are then subjected to the solderability test (see 3.8). No visual defects are allowed.

(3) Remaining three samples are subjected to the resistance to soldering heat test (see 3.20).

(4) All six samples shall be subjected to groups A2 and A4 inspections. Minor scratching of the terminals due to insertion into test sockets shall not be cause for rejection.

3.5.7.4.2 Solder dip/retinning options. The manufacturer may solder dip/retin as follows:

a. After the 100 percent group A1 screening tests and before the group A3 solderability test. Following the solder dip/retinning process, insulation resistance (all terminals to case), seal, and visual/mechanical inspection shall be performed.

b. As a corrective action, if the lot fails the group A solderability test, insulation resistance (all terminals to case), seal, and visual/mechanical inspection shall be performed..

c. For relays that have been subjected to and passed group A inspection. Following the solder dip/retinning process, insulation resistance (all terminals to case), seal, and visual/mechanical inspection shall be performed.

3.5.8 Diodes. Relays supplied with diodes installed internally are not considered electrostatic discharge (ESD) sensitive. However, the diode may be ESD sensitive when not part of the coil circuit or wired internal to the coil. In such case, the diode shall be processed in accordance with the requirements specified in 4.1.4. Manufacturers may, at their option, test diodes used internally as specified in method 3015 of MIL-STD-883 modified to 16,000 volts to eliminate the need for the ESDS protection program described above.

3.6 In-process inspection (see 4.7.1).

3.6.1 Diode in-process screening (see 4.7.1.1). Perform in-process screening as specified in 4.7.1.1.

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3.6.2 Cleaning and small particle inspection (when specified, see 3.1) (see 4.7.1.2). Prior to hermetic sealing of the relay header to the relay can; the relays shall be cleaned and examined for small particle contaminants as specified in 4.8.2.

3.7 Screening (see 4.8.3). The contact miss detector's monitoring level shall be less than 100 ohms for relays tested during cycling. Unless otherwise specified (see 3.1), any relay shall have a final insulation resistance measurement of 10,000 megohms or greater.

3.8 Solderability (see 4.8.4). The critical (examination) area of solid wire lead and pin terminals shall be at least 95 percent covered with a continuous new solder coating per method 208 of MIL-STD-202. For solder-lug terminals greater than .045 inch (1.14 mm) in diameter, 95 percent of the total length of fillet, which is between the standard wrap wire and the terminal, shall be tangent to the surface of the terminal being tested, and shall be free of pinholes, voids, etc. A ragged or interrupted line at the point of tangency between the fillet and the terminal under test shall be considered a failure.

3.9 Seal (see 4.8.5). There shall be no leakage in excess of 1×10^{-8} atmospheric cubic centimeters per second of air (atm cm3/s).

3.10 Insulation resistance (see 4.8.6). The insulation resistance shall be 10,000 megohms or more, unless otherwise specified (see 3.1). After the high level life tests, the insulation resistance shall be 1,000 megohms or more.

3.11 Dielectric withstanding voltage (see 4.8.7). There shall be no leakage current in excess of 100 microamperes (μA). After high level tests, the dielectric withstanding voltage measured at atmospheric or reduced barometric pressure shall be at least 75 percent of the initial value (see 3.1).

3.12 Electrical characteristics (see 4.8.8). The following tests as specified in 3.12.1 through 3.12.7 shall comprise the electrical characteristics tests. Unless otherwise specified, electrical characteristics shall be 100 percent inspected and performed in the order as shown below.

3.12.1 Static contact resistance (see 4.8.8.1). Unless otherwise specified (see 3.1), the static contact resistance shall not exceed 0.05 ohm.

3.12.2 Specified pickup or latch/reset, hold, and dropout values (voltages) (see 4.8.8.2). The specified pickup or latch/reset, hold, and dropout values (voltages) shall be as specified (see 3.1).

3.12.3 Coil resistance or coil current (see 4.8.8.3).

3.12.3.1 Coil resistance (see 4.8.8.3.1). The coil resistance shall be as specified (see 3.1).

3.12.3.2 Coil current (see 4.8.8.3.2). The coil current shall be as specified (see 3.1) for relays with both coil transient suppression and polarity reversal protection diodes.

3.12.4 Operate and release time (see 4.8.8.4). The operate and release time shall be as specified (see 3.1) In multipole relays, during each of the operate and the release time measurements, the difference between the first moving contact to make and the last moving contact to make shall not exceed 1 millisecond (ms). This shall be exclusive of contact bounce. Release time is not applicable to latching relays.

3.12.4.1 Break before make (see 4.8.8.4.1). Moving contacts within a multipole relay shall show no evidence of any open contact closing before all closed contacts have opened (see 3.1). This applies to either state of the relay.

3.12.5 Contact dynamic characteristics (see 4.8.8.5).

3.12.5.1 Contact bounce (applicable to failure rate level "L") (see 4.8.8.5.1). The duration of the contact bounce shall not exceed 1.5 ms unless otherwise specified (see 3.1).

3.12.5.2 Contact stabilization time (applicable to failure rate levels "M", "P", and "R") (see 4.8.8.5.2). The time to reach and maintain a static contact resistance state shall not exceed 2.0 ms unless otherwise specified (see 3.1)

3 12 6 Coil transient suppression and diode block integrity test (see 4.8 8.6).

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3.12.6.1 Coil transient suppression (applicable to dc operated relays with coil transient suppression diodes) (see 4.8.8.6.1). Coils of dc operated relays shall not generate a back EMF greater than that specified (see 3.1), as maximum induced transient voltage.

3.12.6.2 Diode block integrity test (applicable to dc operated relays with polarity reversal protection diodes) (see 4.8.8.6.2). The leakage current shall not exceed the value specified (see 3.1).

3.12.7 Neutral screen (applicable to latching relays only) (see 4.8.8.7). Latching relays shall be tested as specified in 4.8.8.7.

3.13 Thermal shock (see 4.8.9). Insulation resistance, specified pickup or latch/reset, hold, and must dropout values (voltages), and operate and release time shall meet the requirements of 3.10, 3.12.2, and 3.12.4, respectively, at each temperature extreme. Following the temperature excursions, there shall be no cracking, peeling, or flaking of the finish; dielectric withstanding voltage shall meet 3.11 requirements.

3.14 Shock (specified pulse) (see 4.8.10). Unless otherwise specified (see 3.1), there shall be no opening of closed contacts in excess of 10 microseconds (μs) and there shall be no closure or bridging of open contacts in excess of 1 μs and no evidence of mechanical or electrical damage.

3.15 Vibration (see 4.8.11). Unless otherwise specified (see 3.1), there shall be no opening of closed contacts in excess of 10 μs and there shall be no closure or bridging of open contacts in excess of 1 μs and no evidence of mechanical or electrical damage.

3.16 Acceleration (unless otherwise specified, see 3.1) (see 4.8.12). The contacts of the relay shall remain in the de-energized position with no voltage across the coil and in the energized position when voltage is applied to the coil. Latching relays shall remain in each latched position with no voltage on the coil.

3.17 Terminal strength (see 4.8.13). There shall be no evidence of loosening or breaking of the terminals, nor shall there be any other damage which would adversely affect the normal operation of the relay. Bending of terminals shall not be construed as damage. The glass criteria of 3.30 does not apply here.

3.18 Magnetic interference (when specified, see 3.1) (see 4.8.14). The specified pickup or latch/reset, dropout, and hold values (voltages) shall meet the requirements specified in 3.12.2.

3.19 Coil life (see 4.8.15). There shall be no evidence of damage.

3.19.1 Coil endurance (see 4.8.15.1). The coil resistance shall be as specified (see 3.1).

3.20 Resistance to soldering heat (see 4.8.16). There shall be no damage which would adversely affect normal operation of the relay.

3.21 Salt spray or atmosphere (corrosion) (see 4.8.17). There shall be no evidence of breaking, cracking, chipping, or flaking of the finish, nor exposure of the base metal, due to corrosion, which would adversely affect the application or performance characteristics of the relay.

3.22 Overload (applicable to high level relays only) (see 4.8.18). The voltage drop across closed contacts shall be less than or equal to 5 percent of the applied load voltage and the voltage across open contacts shall be 95 percent or more of the applied load voltage. The case-to-ground fuse shall remain electrically continuous.

3.23 Life (see 4.8.19). For low level testing, the contact miss detector's monitoring level shall be less than or equal to 100 ohms (unless otherwise specified, see 3.1). For high level testing, the contact miss detector's monitoring level shall be less than or equal to 5 percent of the applied load voltage and the voltage across open contacts shall be 95 percent or more of the applied load voltage. Unless otherwise specified, the static contact resistance following cycling shall be no greater than twice the initial specified contact resistance requirement. There shall be no mechanical or electrical failure. Welding of contacts, failure to make, carry or break the load, or failure of the fuse connected between case and load system ground or neutral shall constitute a failure. Relays indicating failure, not verified per a failure verification procedure approved by the Qualifying Activity, may be returned to test. During post life tests, failure of a diode shall constitute a failure.

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3.24 Intermediate current (see 4.8.20). During cycling, unless otherwise specified (see 3.1), the resistance of a closed contact shall be less than or equal to 3 ohms and the voltage across an open contact shall be 95 percent or more of applied load voltage. After cycling, the static contact resistance shall be measured at room ambient (+25°C) and shall not exceed the limits as specified (see 3.1). Intermediate current shall not be considered a low level or high level contact load rating (see 6.1.1). There shall be no mechanical or electrical failure. Welding of contacts, failure to make, carry or break the load, or failure of the fuse connected between case and load system ground or neutral shall constitute a failure. Relays indicating failure, not verified per a failure verification procedure approved by the qualifying activity, may be returned to test. During post life tests, failure of a diode shall constitute a failure.

3.25 Mechanical life (see 4.8.21). After cycling, the insulation resistance and dielectric withstanding voltage shall not exceed the limits as specified (see 3.1) and the operate and release time shall not exceed 120 percent of the limits specified (see 3.1). There shall be no mechanical or electrical failure. Welding of contacts, failure to make, carry or break the load, or failure of the fuse connected between case and load system ground or neutral shall constitute a failure. Relays indicating failure, not verified per a failure verification procedure approved by the qualifying activity, may be returned to test. During post life tests, failure of a diode shall constitute a failure.

3.26 Resistance to solvents (see 4.8.22). The marking shall remain legible.

3.27 Particle impact noise detection (PIND, when specified, see 3.1) (see 4.8.23). There shall be no evidence of free moving particulate contamination.

3.28 Marking.

3.28.1 "JAN" brand. The United States Government has adopted, and is exercising legitimate control over the certification marks "JAN" and "J", respectively, to indicate that items so marked or identified are manufactured to, and meet all the requirements of military specifications. Accordingly, items acquired to, and meeting all of the criteria specified herein and in applicable specifications shall bear the certification mark "JAN" except that items too small to bear the certification mark "JAN" shall bear the letter "J". The "JAN" or "J" shall be placed on the first line above or below the "M" of the military part number or the "J" with the date code (example J8530). Items furnished under contracts or orders which either permit or require deviation from the conditions or requirements specified herein or in applicable specifications shall not bear "JAN" or "J". In the event an item fails to meet the requirements of this specification and the applicable specification sheets or associated detail specifications, the manufacturer shall remove the "JAN" or the "J" from the sample tested and also from all items represented by the sample. The "JAN" or "J" certification mark shall not be used on products acquired to contractor drawings or specifications. The United States Government has obtained Certificate of Registration No. 504,860 for the certification mark "JAN".

3.28.2 Identification marking (full). Relays shall be marked in accordance with method I of MIL-STD-1285 and shall include the following information:

a. Military PIN (see 1.2 and 3.1). The "JAN" or "J" shall not be marked in front of the PIN.

b. "JAN" or "J" brand. The "JAN" or "J" shall appear directly above or below the "M" of the military PIN (examples: JAN M39016/6-109L
M39016/6-109L or J

c. Date code (at the option of the manufacturer, the "J" with the date code may be used instead of b). The date code shall provide traceability (see 4.1.2.1).

d. Source code.

e. Lot symbol (optional).

f. Rated coil voltage (or current) (see 3.1) and when applicable, operating frequency.

g. Coil resistance.

h. Contact rating (the highest dc resistive load rating shall be marked) (see 3.1).

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i. Circuit diagram (see 3.5.5).

j. Terminal marking (when applicable, see 3.1, 3.5.4.1, and 3.5.7).

3.28.3 Minimum marking (when specified, see 3.1). When space does not permit the marking specified in 3.28.2, the marking shall include, as a minimum the military PIN; the "J" with the date code (example J8530); circuit diagram; and manufacturer's name or source code.

3.28.4 Failure rate level substitution. With procuring activity approval, relays qualified to lower (better) failure rates, may be substituted for higher failure rate parts. For example, a relay qualified to failure rate level "P" (0.1 percent/10,000 cycles), may be substituted for a failure rate "M" (1.0 percent/10,000 cycles), relay. Relays shall not be remarked unless specified in the contract or purchase order.

3.28.5 Interchangeability. All parts having the same military PIN shall be directly and completely interchangeable with each other with respect to installation and performance to the extent specified in the military specification sheet (see 3.1).

3.29 Workmanship. The relays shall be fabricated in such a manner as to be uniform in quality, and shall be free from cracked or displaced parts, sharp edges, burrs, and other defects that will affect life, serviceability, and appearance.

3.30 Header glass. Header glass may have small irregularities, such as bubbles, chips, and cracks. The acceptability shall be based on figure 2 and the following:

a. Surface blisters whose diameters do not exceed one-half of the radial distance between the terminal and the corresponding header metal (for a cluster of blisters, the overall cluster diameter shall apply) are acceptable.

b. Circumferential cracks which originate and terminate in the glass and which extend less than 135° are acceptable (see figure 2, reference B).

c. Radial cracks with lengths not exceeding three-quarters of the distance between the terminal and the surrounding header metal are acceptable (see figure 2, reference C).

d. Tangential cracks which are confined to a single zone are acceptable (see figure 2, reference D).

e. Meniscus cracks, and chipouts, which completely surround the terminal are acceptable if:

(1) The meniscus cracks, and chipouts do not extend beyond one-half the distance from the terminal to the surrounding header metal, or

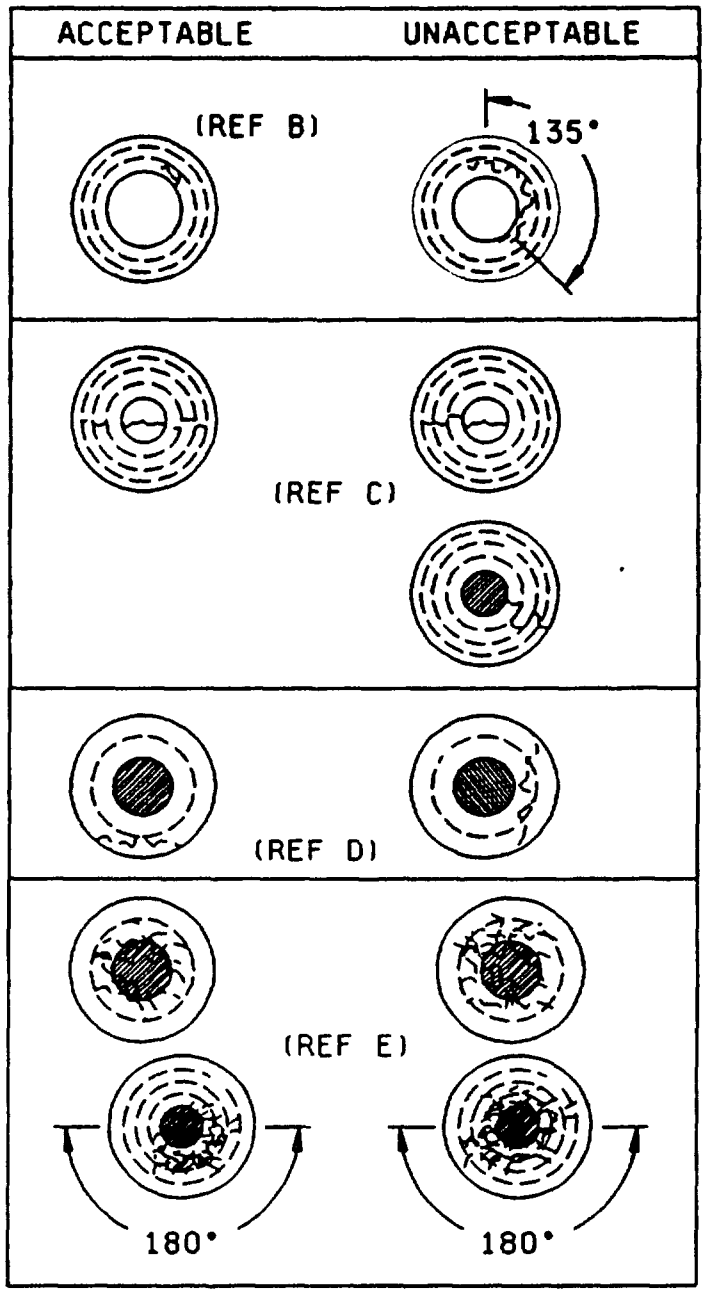
(2) The meniscus cracks, and chipouts do not extend beyond one-quarter of the distance from the terminal to the surrounding header metal and a portion of the meniscus cracks, and chipouts does not extend beyond three-quarters of the distance from the terminal to the surrounding header metal if that portion of the meniscus cracks, and chipouts does not span more the 180° (see figure 2, reference E).

f An internal bubble whose diameter does not exceed one-third the distance between the terminal and the surrounding header metal is acceptable.

Microscopic examination with up to 10X magnification shall be used. In case of dispute, all relays shall meet the applicable insulation resistance, dielectric withstanding voltage, and seal requirements, regardless of the acceptability of the header glass.

The above requirements are not applicable after group B and group C inspections and also not applicable to relays with spacer pads after forming the leads.

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NOTE: The broken lines indicate the division of the radial distance between the terminal and the surrounding header metal into equal parts, thus creating concentric zones around the terminal

FIGURE 2 Inspection aid

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and service conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Product assurance program. A product assurance program shall be established and maintained in accordance with MIL-STD-790. Evidence of such compliance shall be verified by the qualifying activity of this specification as a prerequisite for qualification and continued qualification.

4.1.2 1 Traceability requirements. The manufacturer shall submit to the qualifying activity the procedure whereby the lot date codes are assigned that incorporates traceability. The following is a list of raw material/component parts and subassembly traceability requirements:

a. Header-contact subassembly with the lot number.

(1) Stationary or moving contact (a contact may consist of an individual wire or a contact blade and contact button).

(2) Header with glass to metal sealed leads in place with the lot number.

(3) Return spring(s).

(4) Diodes (when applicable) with the lot number.

(5) Magnets (when applicable).

b. Motor subassembly with the lot number.

(1) Wound coils with the lot number

(2) Armature assembly (including actuators as applicable).

(3) Diodes (when applicable) with the lot number.

(4) Magnets (when applicable).

4.1.3 Statistical process control (SPC). A SPC program shall be established and maintained in accordance with EIA-557. Evidence of such compliance shall be verified by the qualifying activity as a prerequisite for qualification and retention of qualification.

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4.1.4 ESD protection program. This requirement is applicable to all manufacturers who handle ESDS component parts and/or materials in the relay manufacturing and /or testing process. The manufacturer shall establish and maintain an ESD control program in accordance with MIL-STD-1686 and MIL-HDBK-263. Evidence of such compliance shall be verified by the qualifying activity of this specification as a prerequisite for qualification and continued qualification. This program shall be documented by an ESD control plan which must be under document control. As a minimum, this plan must address the identification of ESDS sub-components and end items, facilities, training, design protection, handling procedures, marking, cleaning, preservation, packaging, and quality assurance. A model ESD control program is available from the qualifying activity and may be used as a guideline document. Further guidance for ESD control is available from the EOS/ESD Association and the Electronics Industry Association (EIA).

4.2 Classification of inspection. The inspections specified herein are classified as follows:

a. Qualification inspection (see 4.4).

b. Verification of qualification (see 4.5).

c. In-process and quality conformance inspection (see 4.7).

4.2.1 Records retention. Documentation for test and production shall be retained as a minimum as defined below:

a. Qualification test reports, data, and samples shall be retained as long as qualification is maintained.

b. Group A test data and production traveler: 3 years from the final inspection or date code.

c. Group B and C test reports and data: 5 years from date of submission. Test samples shall be retained until a letter of acceptance has been received from the qualifying activity or for one reporting period, whichever is longer.

4.3 Inspection conditions. Unless otherwise specified herein, the test conditions specified in the "GENERAL REQUIREMENTS" of MIL-STD-202 shall be considered for referee purposes only. All inspections may be performed at ambient conditions consistent with industry practice.

4.3.1 Power supply. Unless otherwise specified herein, the power supply shall have no more than 10 percent regulation at 110 percent of the specified test load current. A dc power supply shall have no more than 5 percent ripple voltage. An ac power supply shall be within 1 percent of the specified frequency and shall be sinusoidal with a form factor between 0.95 and 1.25.

4.3.2 Grounding. Unless otherwise specified (see 3.1), the negative side of the dc power supply shall be grounded. One side of single-phase ac power supply shall be grounded.

4.3.3 Load conditions during tests. The coil(s) of the relay being tested shall be connected to a power supply separate from the load power supply. The loads of the relay being tested shall be connected individually to each stationary contact. The polarity of the load power supply is optional. The movable contacts of the relay being tested may be connected to either the high or low side of the load power supply. All tests during which the contacts are loaded and being cycled, except dielectric withstanding voltage, shall be conducted with the case of the relay connected to the power supply ground or neutral through a normal blow fuse rated at 5 percent of the contact load maximum, but not less than 0.100 ampere. For relays with nongrounded case ratings, tests for isolated-case ratings may be made with the case electrically isolated from the power supply ground.

4.3.4 Testing devices. Devices used in the testing of relays shall not load the contacts above 10 mA resistive at 6 V dc or peak ac maximum open circuit voltage unless otherwise specified herein.

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4.3.5 Mounting relays for ambient temperature tests. When the relays are subjected to the tests specified in 4.8.20 and 4.8.15 (intermediate current and coil life, respectively), they may be mounted on a heat sink in accordance with the following:

a. Each relay may be attached by its normal mounting means to a .062 inch (1.57 mm) thick minimum, flat aluminum plate heat sink. The heat sink shall be designed to place every relay in the center of its own square space whose total surface area (both sides) is eight times the outside surface area of the relay, excluding mounting. Relays without mounts shall be held to the heat sink with a metal strap .25 inch (6.35 mm) wide by .015 inch (0.38 mm) maximum thickness. The heat sink assembly shall be suspended by twine or other nonheat conducting material. The leads shall not constitute a heat sink.

b. Chamber temperature shall be controlled to maintain the temperature at the specified ambient extremes (see 3.1).

4.3.6 Methods of examination and test. Application of coil power to relays under test shall be such that plus polarity is applied to the color coded terminal when applicable; or to the lower numbered terminal when color coding is not used. Testing of latching relays shall be repeated with each coil serving as the operating coil; and testing of latching relays shall be repeated with the relay in each operated position.

4.3.7 Reliability requirements. The reliability requirements specified herein are in accordance with MIL-STD-690. The confidence level for qualification is 90 percent and the confidence level for maintenance of qualification is 60 percent.

4.3.8 Tolerances. Unless otherwise specified (see 3.1), all electrical, environmental, and mechanical parameters shall have a tolerance of ±10 percent.

4.3.9 Alternate test equipment. Test circuits and test equipment herein are intended to provide guidance to the relay manufacturer. Use of any alternate test circuits and/or test equipment shall be approved by the qualifying activity prior to use.

4.4 Qualification inspection. Qualification inspection shall be performed at a laboratory acceptable to the Government (see 6.3) on sample units produced with equipment and procedures normally used in production.

4.4.1 Qualification. Qualification shall be granted at the "L" or "M" failure rate initially and shall be based on results of the qualification inspection specified in table II. A certification of construction to the materials requirements of 3.4, and the design and construction requirements of 3.5 shall accompany the submission of qualification inspection results to the qualifying activity.

4.4.1.1 Sampling plan. The number of relays to be subjected to qualification inspection shall be as specified in table II. The sample shall be selected from a production run and shall be produced with equipment or procedures normally used in production. The qualification sample shall be as defined in table II.

4.4.2 Inspection routine. Sample units shall be subjected to the qualification inspection outlined in table II, in the order shown, except that groups Q3 through Q8 may be conducted concurrently. All sample units shall be subjected to the tests in Q2. These sample units shall then be divided into six groups as specified in table II and subjected to the inspection specified for their particular group.

4.4.3 Failures. Failures in excess of those allowed in table II shall be cause for refusal to grant qualification approval.

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TABLE II. Qualification inspection.

Inspection	Requirement paragraph	Test method paragraph	Number of sample units to be inspected	Number of failures allowed
Q1				
Cleaning and small particle inspection (when specified)	3.6.2	4.8.2	All sample units	1/
Q2				
Visual and mechanical inspection (internal) 2/	3.1 and 3.29	4.8.1		
Screening	3.7	4.8.3		
Solderability (3 sample units) 3/	3.8	4.8.4		
Dielectric withstanding voltage 4/	3.11	4.8.7	All sample units	0
Insulation resistance 4/	3.10	4.8.6		
Electrical characteristics 4/5/	3.12	4.8.8		
Visual and mechanical inspection (external) (dimensional check on 2 sample units only)	3.1, 3.28, 3.29, 3.30	4.8.1		
Seal	3.9	4.8.5		
Q3				
Thermal shock	3.13	4.8.9		
Resistance to solvents	3.26	4.8.22		
Shock (specified pulse)	3.14	4.8.10		
Vibration (sinusoidal)	3.15	4.8.11.1	4	1
Vibration (random)	3.15	4.8.11.2		
Particle impact noise detection (PIND, when specified)	3.27	4.8.23		
Acceleration	3.16	4.8.12		
Terminal strength	3.17	4.8.13		
Dielectric withstanding voltage 4/	3.11	4.8.7		
Insulation resistance 4/	3.10	4.8.6		
Electrical characteristics 4/ 5/	3.12	4.8.8		
Seal	3 9	4.8.5		
Q4				
Magnetic interference (when specified)	3.18	4.8.14		
Coil life	3 19	4.8.15		
Resistance to soldering heat	3.20	4.8.16		
Salt atmosphere (corrosion)	3.21	4.8.17		
Dielectric withstanding voltage 4/	3.11	4.8.7	4	1
Insulation resistance 4/	3.10	4.8.6		
Electrical characteristics 4/ 5/	3.12	4.8.8		
Seal	3.9	4.8.5		

See footnotes at end of table.

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TABLE II. Qualification inspection - Continued.

Inspection	Requirement paragraph	Test method paragraph	Number of sample units to be inspected	Number of failures allowed
Q5				
Overload (highest dc resistive load)	3.22	4.8.18		
Life (highest rated dc, 50,000 cycles) 6/	3.23	4.8.19	4	1
Dielectric withstanding voltage 4/	3.11	4.8.7		
Insulation resistance 4/	3.10	4.8.6		
Electrical characteristics 4/ 5/	3.12	4.8.8		
Q6			For qualification level L	
			8 -	- - 0
Life 7/	3.23	4.8.19	14 -	- - 1
Dielectric withstanding voltage 4/	3.11	4 8.7	23 -	- - 2
Insulation resistance 4/	3.10	4.8.6	40 -	- - 5
Electrical characteristics 4/ 5/	3.12	4.8.8	For qualification level M	
Q7			23 -	- - 0
			39 -	- - 1
Intermediate current	3.24	4.8.20		
Dielectric withstanding voltage 4/	3 11	4.8.7		
Insulation resistance 4/	3.10	4.8.6	4	1
Electrical characteristics 4/ 5/	3.12	4.8.8		
Q8				
Mechanical life	3.25	4.8.21		
Dielectric withstanding voltage	3.11	4.8.7		
Insulation resistance	3 10	4 8.6	4	1
Operate and release time	3.12.4	4.8.8.4		
Visual inspection (external)	3.1,3.28,3.29	4.8.1		

1/ Cleaning and small particle inspection shall not contribute failures for qualification purposes. Upon completion of cleaning and small particle inspection the qualification samples shall continue through manufacturing prior to entrance into Q2 through Q8. Manufacturers should select a sufficiently large enough sample size to insure the relays required for the sample size specified in Q6 are available upon completion of all manufacturing processes prior to beginning of the testing specified in Q2 through Q8.

2/ One sample unit remaining unsealed shall be subjected to the internal inspection

3/ Solderability samples are not subjected to dielectric withstanding voltage, insulation resistance, electrical characteristics, and seal.

4/ Coil resistance or coil current may be done prior to dielectric withstanding voltage and insulation resistance. Testing sequence is optional for insulation resistance and dielectric withstanding voltage prior to electrical characteristics.

5/ Measurement of coil current is applicable only to relays with both coil transient suppression and polarity protection diodes.

6/ Post overload life testing requirements are not applicable to TO-5 and 0.100 grid style relays.

7/ The sample size shall be equally divided among the specified contact ratings and shall be of sufficient size to test a minimum of one relay per contact rating with rated loads on all contacts When there are more samples than loads, the remaining loads shall be assigned in sequence beginning with the highest dc rated load.

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4.4.4 Failure-rate level and quality level verification.

4.4.4.1 FR qualification. FR qualification shall be in accordance with the general and detailed requirements of MIL-STD-690, MIL-STD-790 in its entirety, and the following details:

a. Procedure I - Qualification at the initial FR level. The manufacturer may qualify his products to failure rate levels "L" or "M" initially.

b. Procedure II - Extension of qualification to lower FR levels. To extend qualification to a lower FR level (see table III), data from two or more styles of similar construction may be combined.

c. Procedure III - Maintenance of FR level qualifications. Maintenance period of table IV shall apply. Regardless of the number of production lots produced during this period, the specified number of life test unit cycles shall be accumulated to maintain qualification. If there is no production, see 4.5. The manufacturer shall preselect the sampling plan to be used during the maintenance period from table IV. In the event that failures occur exceeding the sampling plan selected (C = number), the failure rate of the product shall be increased one level or the product removed from the Qualified Products List. Noncompliance corrective action shall be in accordance with 4.7.3.3.

4.5 Verification of qualification. At all failure rate levels and maintenance periods specified in table IV, the manufacturer shall compile a summary of the results of quality conformance inspections and (where applicable) extended FR test data, in the form of a verification of qualification report, and forward it to the qualifying activity as the basis of continued qualification approval at 6-month intervals. In addition to the periodic submission of FR test data, the manufacturer shall immediately notify the qualifying activity whenever the FR data indicates that the manufacturer has failed to maintain his qualified FR level. Continuation shall be based on evidence that, over the 6-month period, the following has been met:

a. Verification by the qualifying activity that the manufacturer meets the requirement of MIL-STD-790.

b. The manufacturer has not modified the design of the item.

c. The specification requirements for the item have not been amended so as to affect the character of the item.

d. Lot rejection for group A inspection does not exceed 10 percent or one lot, whichever is greater.

e. Requirements for groups B and C are met (where applicable).

f. The records of all FR tests combined substantiate that the "L" (3.0 percent), "M" (1.0 percent), or "P" (0.1 percent) FR levels have been maintained or that the manufacturer continues to meet the "R" (0.01 percent) FR level for which qualified, although the total component cycles of testing do not, as yet, meet the requirements of 4.4.4 (see table IV).

If group C test requirements were not met and the manufacturer has taken corrective action satisfactory to the Government, the forwarding of the verification of qualification report may be delayed until within 60 days after completion of retesting of the group C inspections. In this case, the qualifying activity shall be notified of this condition within the time the original verification of qualification report was due. All reports shall be certified by a responsible company official and the Government inspector. In the event that no production occurred during a 6-month reporting period, the manufacturer shall submit a DD Form 1718, Certification of Qualified Product, to the qualifying activity certifying that the company still has the capabilities and facilities necessary to produce the relay. If during two consecutive reporting periods there has been no production, the manufacturer may be required, at the discretion of the qualifying activity, to produce test samples to perform all group B and C tests.

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TABLE III. Requirements for extension of qualification to lower (better) failure rate levels (90 percent confidence level).

FR level symbol	Qualified FR (%/10,000 cycles)	Cumulative unit cycles in millions					
		C 1/ = 0	C = 1	C = 2	C = 3	C = 4	C = 5
M	1.0	2.30	3.89	5.32	6.68	7.99	9.27
P	0.1	23.0	38.9	53.2	66.8	79.9	92.7
R	0.01	230	389	532	668	799	927

1/ C = Acceptance number or number of failures permitted. C numbers greater than five shall be coordinated and approved by the qualifying activity.

TABLE IV. Sampling plan for maintenance of FR level qualification (60 percent confidence level).

FR level symbol	Qualified FR (%/10,000 cycles)	Maximum qualification maintenance period	Cumulative unit cycles in millions				
			C 1/ = 1	C = 2	C = 3	C = 4	C = 5
L	3.0	6 months	.673	1.03	1.39	1 75	2.10
M	1.0	6 months	2.02	3.10	4.18	5.25	6.30
P	0.1	12 months	20.2	31.0	41.8	52.5	63.0
R	0.01	24 months	202	310	418	525	630

1/ C = Acceptance number or number of failures permitted. C numbers greater than five shall be coordinated and approved by the qualifying activity.

4.6 FR level determination. Determination of FR levels shall be based upon data from all completed life tests. Data for determination of FR levels shall be accumulated in the following manner:

a. The qualification test samples.

b Test on every production lot which has been submitted for group B quality conformance inspection.

c. Test results of relays which have a similar internal design and construction, same production processes, same or higher environmental capability, and same electrical characteristics as the qualified relay and which have been tested in accordance with the group B quality conformance inspection conditions. These relay types must be reviewed and approved by the qualifying activity prior to being considered as acceptable relays for use in obtaining FR data.

4.6.1 Records. Test records shall be maintained which shall include the data derived from the sources specified in 4.6. The example forms shown in MIL-STD-690 include the minimum information required and a suggested format for this report. Test data on every production lot that has been submitted for quality conformance inspection shall be included.

4.6.2 FR level computation. Data from test records shall be used to compute the observed failure rates of the sample units using a maintenance of FR record form similar to examples of FR test records (figure 1) of MIL-STD-690. For this computation, a sample unit which fails at any given cycle shall be classed as a failure for all subsequent measurements. However, the contractor, at his option, may physically remove failed sample units from the test If a failed sample unit is not removed, the cycles accumulated subsequent to its failure shall not be counted toward the cumulative component cycles shown on record form.

4.6.2.1 Contact FR computation The data from tests shall be recorded in such a manner so that, if necessary, it can be analyzed to compute contact FR

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4.6.3 Exemption of data. Exemptions of data requirement in MIL-STD-690 shall apply.

4.6.4 Qualification approval for higher FR. Qualification approval granted for one of the lower FR levels will include qualification for all of the higher FR levels; e.g., qualification approval for level "R" will include qualification approval for levels "P", "M", and "L".

4.7 In-process and quality conformance inspection.

4.7.1 In-process inspection (see 3.6).

4.7.1.1 Diode in-process screening (see 3.6.1). Each manufacturer shall establish a diode screening process as approved by the qualifying activity. As a minimum, this process shall include a 24-hour minimum burn-in at +125°C minimum and may use MIL-STD-750 as a guide. This process shall be performed prior to run-in (see 4.8.3.2).

4.7.1.2 Cleaning and small particle inspection (when specified) (see 3.6.2). This testing shall be 100 percent inspection and shall be done in-process in accordance with 3.6.2 and 4.8.2.

4.7.2 Inspection of product for delivery. Inspection of product for delivery shall consist of group A inspection.

4.7.2.1 Inspection and production lot.

4.7.2.1.1 Inspection lot. An inspection lot shall consist of all the relays of the same specification sheet, and those of similar design and construction as outlined in 6.3.2, manufactured under essentially the same process and conditions during a manufacturing period of 1 month maximum. The manufacturer shall define and document the period for an inspection lot. For purposes of lot formation, all terminal types may be included in the same lot; however, all lead types which are combined shall have the same method of terminal attachment. All leads in the lot shall be represented in a similar proportion by samples selected for inspection.

4.7.2.1.2 Production lot. A production lot shall consist of all relays of the same PIN. Manufacture of all parts in the lot shall have been started, processed, assembled, and tested as a group. Lot identity shall be maintained throughout the manufacturing cycle.

4.7.2.2 Group A inspection. Group A inspection shall consist of the inspections specified in table V.

4.7.2.2.1 A1, A2, and A4 tests. The A1, A2, and A4 tests shall be performed on a production lot basis on 100 percent of the product supplied under this specification. Relays that do not meet specification requirements of these groups shall be removed from the lot.

4.7.2.2.1.1 Sampling plan. The tests in A1, A2, and A4 shall be performed on each relay offered for inspection, except as noted. When possible, these tests shall be witnessed by the government inspector at the time they are performed by the contractor.

4.7.2.2.1.2 Rejected lots. Relays that do not meet the requirements of the tests in A1 and A4 shall be rejected and shall be removed from the lot. If more than 5 percent of the relays are discarded during A2 testing, the production lot shall be rejected and not offered for reinspection.

4.7.2.2.2 A3 tests.

4.7.2.2.2.1 Sampling plan. Two samples shall be selected randomly from each inspection lot and subjected to the A3 solderability test. If there are one or more defects, the lot shall be considered to have failed. Relays failing the A2 electrical tests, or rejected for other criteria which will not influence solderability, may be used for solderability testing. All rejected relays used as samples must accompany the acceptable relays through all processing environments. The rejected relays shall be marked in a definite manner in order to preclude mixing with acceptable parts.

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TABLE V. Group A inspection.

Inspection	Requirement paragraph	Test method paragraph	Inspection requirements
A1 1/			
Vibration (sinusoidal)	3.15	4.8.11.1	
Vibration (random) 2/	3.15	4.8.11.2	
Particle impact noise detection (PIND, when specified)	3.27	4.8.23	100 percent
Screening, internal moisture	3.7	4.8.3.1	
Screening, run-in	3.7	4.8.3.2	
A2			
Dielectric withstanding voltage 3/	3.11	4.8.7	
Insulation resistance 3/	3.10	4.8.6	100 percent
Electrical characteristics 3/ 4/	3.12	4.8.8	
A3			
Solderability	3.8	4.8.4	2 samples
A4			
Visual and mechanical inspection (external) 5/ 6/	3.1, 3.28, 3.29, 3.30	4.8.1	100 percent
Seal	3.9	4.8.5	

1/ Tests shall be performed in the order shown.

2/ Random vibration shall be performed when specified (see 3.1).

3/ Coil resistance or coil current may be done prior to dielectric withstanding voltage and insulation resistance. Testing sequence is optional for insulation resistance and dielectric withstanding voltage prior to the electrical characteristics.

4/ Measurement of coil current is applicable only to relays with both coil transient suppression and polarity reversal protection diodes.

5/ Minor defects, such as marking, may be reworked.

6/ It is intended that a two piece sample of each relay style be inspected in accordance with 6.3.2, except as limited by the housing style, external attachments, etc..

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4.7.2.2.2.2 Rejected lots. In the event of one or more defects, the inspection lot is rejected. The manufacturer may use one of the following options to rework the lot:

a. Each production lot that was used to form the failed inspection lot shall be individually submitted to the solderability test as required in 4.8.4. Production lots that pass the solderability test are acceptable for shipment. Production lots failing the solderability test can be reworked as described in 4.7.2.2.2.2b and 4.7.2.2.2.2c.

b. The manufacturer submits the failed lot to a 100 percent solder dip using an approved solder dip process in accordance with 3.5.7.4. Two additional samples shall be selected and subjected to the solderability test with zero defects allowed. If the lot fails this solderability test, the lot shall be rejected and shall not be furnished against the requirements of this specification.

c. The manufacturer may submit the failed lot to a strip and replate process followed by a complete group A inspection. The reworked lot shall be considered a new lot for the purpose of the 5 percent requirement of A2. If the lot fails this solderability test, the lot shall be rejected and shall not be furnished against the requirements of this specification.

4.7.3 Periodic inspections. Periodic inspections shall consist of groups B and C. Except where the results of these inspections show noncompliance with the applicable requirements (see 4.7.3.3), delivery of products which have passed group A shall not be delayed pending the results of these periodic inspections.

4.7.3.1 Group B inspection. Group B inspection shall consist of the tests in the order specified in table VI and shall be made on sample units from inspection lots which have passed the group A inspection.

4.7.3.1.1 Sampling plan. The test sample size shall be determined by the manufacturer so that the relay cycles generated meet the maintenance of qualification requirements specified for the qualified failure rate level (see 4.4.4). A minimum quantity of samples shall be selected from each inspection lot as determined by the quantity needed to maintain the FR level divided by the number of inspection lots in the reporting period. In all cases, the sampling plan shall be approved by the qualifying activity. As far as is practicable, the relays tested during a maintenance period shall be representative of all relays produced during this period. The accumulated data shall be used for maintenance and extension of failure rate qualification.

4.7.3.1.2 Disposition of sample units. Sample units which have been subjected to group B inspection shall not be delivered on the contract or purchase order.

TABLE VI. Group B inspection.

Inspection	Requirement paragraph	Test method paragraph
Life 1/	3.23	4.8.19
Dielectric withstanding voltage 2/	3.11	4.8.7
Insulation resistance 2/	3.10	4.8.6
Electrical characteristics 2/ 3/	3.12	4.8.8
Visual inspection (external)	3.1, 3.28, 3.29	4.8.1

1/ Sample units shall be tested in a predetermined sequence as defined by the manufacturer and approved by the qualifying activity. All loads specified in the applicable specification sheet shall be represented. All loads need not be represented in a 6-month maintenance period but shall be represented within two 6-month maintenance periods.

2/ Coil resistance or coil current may be performed prior to dielectric withstanding voltage and insulation resistance. The testing sequence is optional for insulation resistance and dielectric withstanding voltage prior to the electrical characteristics.

3/ Measurement of coil current is applicable only to relays with both coil transient suppression and polarity reversal protection diodes.

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4.7.3.2 Group C inspection. Group C inspection shall consist of the tests in the order specified in table VII. Group C inspection shall be made on sample units selected from inspection lots which have passed the group A inspection.

4.7.3.2.1 Sampling plan. Three sample units shall be taken from production every month for C1, two samples for C2 every 6 months, two samples for C3 every 6 months, two samples for C4 every 6 months, two samples for C5 every 12 months, and two samples for C6 every 36 months.

4.7.3.2.1.1 Similarity. For purposes of complying with group C requalification, relays, of similar construction and processing may be combined (see 6.3.2.1 and 6.3.2.2).

4.7.3.2.2 Failures. No failures shall be permitted except for intermediate current in C1 1/.

4.7.3.2.3 Disposition of sample units. Sample units which have been subjected to group C inspection shall not be delivered on the contract or purchase order.

4.7.3.3 Noncompliance. During group B and C inspections, when a failure exceeds the allowable number of failures, the manufacturer shall notify the qualifying activity and the cognizant inspection activity of such failure within 5 working days. The manufacturer shall prepare to take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same materials and processes, and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the qualifying activity has been taken. After the corrective action has been taken, group B and C inspections shall be repeated on additional sample units (all tests, or the tests which the original sample failed, at the option of the qualifying activity). Group A inspection may be reinstituted; however, final acceptance and shipment shall be withheld until the group B or C inspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure shall be furnished to the qualifying activity and the cognizant inspection activity within 5 working days.

4.7.4 Inspection of packaging. The sampling and inspection of the preservation and interior pack marking shall be in accordance with the group A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.8 Methods of inspection.

4.8.1 Visual and mechanical. Relays shall be examined to verify that the materials, external design and construction, physical dimensions, marking, and workmanship are in accordance with the applicable requirements (see 3.1, 3.4, 3.5, 3.28, and 3.29). Paragraph 3.30 inspection requirements apply only when indicated.

4.8.2 Cleaning and small particle inspection (see 3.6). Prior to hermetic sealing of the relay, a qualification activity approved procedure for cleaning and small particle inspection shall be used. Appendix A may be used as a guideline.

4.8.3 Screening (see 3.7).

4.8.3.1 Internal moisture Relays (coils de-energized) shall be held at +15°C to +25°C for a minimum of 30 minutes. The insulation resistance of all contact pins to case only, shall be measured and observed. The relay coil shall be energized with 140 percent of rated voltage for a period of 2 minutes 30 seconds. The insulation resistance of all contact pins to case only shall be verified a minimum of once each 30 seconds during this period and the lowest value shall meet the requirements of 3.7.

1/ One intermediate current failure per 6-month retention period shall be permitted per FR groupings of similar relay types as approved by the qualifying activity.

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TABLE VII. Group C inspection.

Inspection	Requirement paragraph	Test method paragraph	Number of sample units for inspection
C1 (every month) 1/			
Intermediate current	3.24	4.8.20	
Dielectric withstanding voltage 2/	3.11	4.8.7	
Insulation resistance 2/	3.10	4.8.6	3
Electrical characteristics 2/ 3/	3.12	4.8.8	
Visual inspection (external)	3.1, 3.28, 3.29	4.8.1	
C2 (every 6 months) 4/			
Thermal shock	3.13	4.8.9	
Shock (specified pulse) 5/	3.14	4.8.10	
Vibration 5/	3.15	4.8.11	
Terminal strength	3.17	4.8.13	2
Dielectric withstanding voltage 2/	3.11	4.8.7	
Insulation resistance 2/	3.10	4.8.6	
Electrical characteristics 2/ 3/	3.12	4.8.8	
Visual inspection (external)	3.1, 3.28, 3.29	4.8.1	
Seal	3.9	4.8.5	
C3 (every 6 months)			
Overload (highest dc resistive load)	3.22	4.8.18.1	
Life (highest rated dc, 50,000 cycles) 6/	3.23	4.8.19.1	2
Dielectric withstanding voltage 2/	3.11	4.8.7	
Insulation resistance 2/	3.10	4.8.6	
Electrical characteristics 2/ 3/	3.12	4.8.8	
Visual inspection (external)	3.1, 3.28, 3.29	4.8.1	
C4 (every 6 months)			
Mechanical life	3.25	4.8.21	
Dielectric withstanding voltage	3.11	4.8.7	2
Insulation resistance	3.10	4.8.6	
Operate and release time	3.12.4	4.8.8.4	
Visual inspection (external)	3.1, 3.28, 3.29	4.8.1	

See footnotes at end of table.

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TABLE VII. Group C inspection - Continued.

Inspection	Requirement paragraph	Test method paragraph	Number of sample units for inspection
C5 (every 12 months) 4/			
Coil endurance	3.19.1	4.8.15.1	
Thermal Shock	3.13	4.8.9	
Vibration (sinusoidal)	3.15	4.8.11.1	
Resistance to soldering heat	3.20	4.8.16	
Dielectric withstanding voltage 2/	3.11	4.8.7	2
Insulation resistance 2/	3.10	4.8.6	
Electrical characteristics 2/3/	3.12	4.8.8	
Visual inspection (external)	3.1, 3.28, 3.29	4.8.1	
Seal	3.9	4.8.5	
C6 (every 36 months) 4/			
Magnetic interference (when specified)	3.18	4.8.14	
Coil life	3.19	4.8.15	
Salt atmosphere (corrosion)	3.21	4.8.17	
Acceleration	3.16	4.8.12	
Resistance to solvents	3.26	4.8.22	2
Dielectric withstanding voltage 2/	3.11	4.8.7	
Insulation resistance 2/	3.10	4.8.6	
Electrical characteristics 2/ 3/	3.12	4.8.8	
Visual inspection (external)	3.1, 3.28, 3.29	4.8.1	
Seal	3.9	4.8.5	

1/ One intermediate current failure per 6-month retention period shall be permitted per FR groupings of similar relay types as approved by the qualifying activity.

2/ Coil resistance or coil current may be performed prior to dielectric withstanding voltage and insulation resistance. The testing sequence is optional for insulation resistance and dielectric withstanding voltage prior to the electrical characteristics.

3/ Measurement of coil current is applicable only to relays with both coil transient suppression and polarity reversal diodes.

4/ Relays with spreader or spacer pads shall not be subjected to the testing of this group.

5/ Each mounting configuration to which a manufacturer is qualified will be represented during the normal continuous periodic testing period. It will usually take more than one periodic testing period to achieve testing of all of the applicable mountings.

6/ Life testing requirements are not applicable to TO-5 and 0.100 grid style relays.

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4.8.3.2 Run-in.

a. High temperature. For qualification inspection only; relays shall be subjected to +125°C with rated voltage or current on the coil circuit for 1 hour, for nonlatching relays; for latching relays, one coil shall be energized for 30 minutes. At the end of this period, the pickup value (voltage), or latch and reset voltage, shall be measured to determine compliance to 3.1. For latching relays, repeat the test after the other coil has been energized for 30 minutes. For group A testing, relays shall be subjected to +125°C; the test chamber shall stabilize at +125°C after the specimens have been inserted into the test chamber. The test shall be performed with rated voltage or current on the coil for 15 minutes minimum for nonlatching relays weighing 1 ounce or less and 30 minutes minimum for relays weighing over 1 ounce but not more than .3 pound. At the end of each period, the specified pickup value (voltage) shall be measured to determine compliance with 3.1. For latching relays, one of the 2 coils shall be energized with rated voltage or current for 7 minutes 30 seconds minimum. For latching relays, repeat this test after the other coil has been energized with rated voltage or current for 7 minutes 30 seconds minimum. The specified pickup value (voltage) shall be measured to determine compliance with 3.1. While at this temperature, the relays shall be subjected to a 2,500 cycle run-in test. The cycling rate shall be calculated using the formulas below and the maximum operate and release times as specified, see 3.1 for the relay under test. For latching relays, use the latch/reset times Relays shall have the contacts loaded as follows: Open circuit load voltage 10 to 50 millivolts (mV) dc or peak ac. The load current shall be 10 to 50 μA. The contact resistance or voltage drop, as applicable, of each pair of mated contacts shall be monitored during 40 percent minimum of each "on" and each "off" period. The test equipment shall automatically turn off when a failure occurs, or shall record every failure.

b. Low temperature. Following high temperature, for qualification only, relays shall be subjected to -65°C with the coil or coils de-energized for 1 hour. For group A testing, relays shall be subjected to -65°C; the test chamber shall stabilize at -65°C after the specimens have been inserted into the test chamber. The test shall be performed with the coil or coils de-energized for 15 minutes minimum for relays weighing 1 ounce or less and 30 minutes minimum for relays weighing over 1 ounce but not more than .3 pound. At the end of either period, the specified dropout value (voltage) shall be measured to determine compliance with 3.1. For latching relays, dropout voltage measurement is not required. While at this temperature, the relays shall be subjected to a 2,500 cycle run-in test in accordance with the procedure outlined in 4.8.3.2a.

$$\text{Maximum cycle rate (cycles per second)} = \frac{0.1}{\text{Maximum operate time (seconds)} + \text{Maximum release time (seconds)}}$$

For latching relays:

$$\text{Maximum cycle rate (cycles per second)} = \frac{0.1}{\text{2X maximum latch/reset time (seconds)}}$$

4.8.4 Solderability (see 3.8). Relays shall be tested in accordance with method 208 of MIL-STD-202. The following detail and exception shall apply: All terminations of each part shall be tested.

4.8.5 Seal (see 3.9). Relays shall be tested in accordance with 4.8.5.1 or 4.8.5.2, as applicable. In case of dispute, method 1014 of MIL-STD-883, test condition B shall govern.

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4.8.5.1 Relays sealed with a tracer gas. Relays sealed with a tracer gas shall be tested in accordance with method 112 of MIL-STD-202, or at the option of the manufacturer, method 1014 of MIL-STD-883. The following details shall apply:

a. Method 112 of MIL-STD-202:

(1) Test condition C, procedure IV. Relays shall be back-filled with a helium tracer gas (90 percent dry gas and 10 percent helium). Silicone oil shall not be used.

(2) Leakage rate sensitivity: 1×10^{-8} atm cm^3/s.

(3) Measurements after test: Not applicable.

b. Method 1014 of MIL-STD-883, test condition B (gross leak test not required).

4.8.5.2 Relays sealed without a tracer gas. Relays sealed without a tracer gas shall be tested in accordance with method 1014 of MIL-STD-883. At the option of the manufacturer, either 4.8.5.2a or 4.8.5.2b may be used. The following details shall apply:

a. Method 1014 of MIL-STD-883:

(1) Test condition A_1 or A_2.

(2) Measurements after test: Perform a gross leak test in accordance with method 112 of MIL-STD-202, test condition A, B, or D. Silicone oil shall not be used. At the option of the manufacturer, the gross leak test of method 1014 of MIL-STD-883, test condition C may be used.

b. Method 1014 of MIL-STD-883, test condition B (gross leak test not required).

4.8.6 Insulation resistance (see 3.10). Relays shall be tested in accordance with method 302 of MIL-STD-202 with the relay in the energized and de-energized (latch/reset) positions. The following details shall apply unless otherwise specified (see 3.1):

a. Test conditions: A (for relays with coil and contact ratings both below 60 volts) and B (for all other relays).

b. Points of measurement: As specified in points of application in table VIII.

c. Attributes data is acceptable.

4.8.7 Dielectric withstanding voltage (see 3.11). Relays shall be tested as specified in 4.8.7.1 and in accordance with 4.8.7.2. Testing in accordance with 4.8.7.2 is not required for group A testing.

4.8.7.1 At atmospheric pressure. Relays shall be tested in accordance with method 301 of MIL-STD-202. The following details shall apply unless otherwise specified (see 3.1):

a. Points of application and magnitude of test voltage: As shown in table VIII.

b. Maximum leakage current: 100 μA.

c. Duration of application: 60 seconds minimum for qualification and groups B and C tests; 5 (60 Hz) cycles minimum for group A tests on test equipment which has an adequate response time to measure the leakage current and is approved by the qualifying activity. For group A tests on other than automatic test equipment (i.e., bench setup), the application time shall be 5 seconds minimum.

d. Attributes data is acceptable.

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TABLE VIII. Test details for dielectric withstanding voltage.

Points of application	Test voltage
Between case, frame, or enclosure and between all contacts in the energized and de-energized positions Between case, frame, or enclosure and coil(s) Between all contacts and coil(s) Between open contacts in the energized and de-energized positions Between coils of latching relays Between contact poles in the energized and de-energized positions (applicable to multipole relays)	1,000 ±5 percent volts ac plus twice rated voltage or the specified value ±5 percent

4.8.7.2 At reduced barometric pressure. Relays shall be tested in accordance with method 105 of MIL-STD-202. The following details shall apply:

a. Method of mounting: Normal mounting means.

b. Test condition: C.

c. Tests during subjection to reduced pressure: As specified in 4.8.7.1, except test voltage shall be 350 V rms unless otherwise specified (see 3.1).

d. Points of application of test voltage: All terminals to case.

e. Duration of application: 60 seconds minimum for qualification and groups B and C tests.

f. Attributes data is acceptable.

4.8.8 Electrical characteristics (see 3.12).

4.8.8.1 Static contact resistance (see 3.12.1). Relays shall be tested in accordance with method 307 of MIL-STD-202. The following details and exception shall apply:

a. Method of connection: For relays with wire-lead terminals, this measurement shall be made .125 inch (3.18 mm) maximum from the emergence of the lead from the relay or bottom of the spreader or spacer pad.

b. Test load: 10 mA maximum at 6 V maximum (dc or peak ac).

c. Post test loads for high level life and intermediate current: Current and voltage shall be the same as the life or intermediate current test or 100 mA maximum at 28 V dc maximum.

d. Points of measurements:

(1) Between all normally closed mated contacts.

(2) Between all normally open mated contacts, with the coil energized with rated coil voltage (or current). Coils shall be de-energized for latching relays.

e. Number of activations prior to measurement: None.

f. Number of test activations: Three.

g Number of measurements per activation: One in each closed contact position.

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4.8.8.2 Specified pickup or latch/reset, hold, and dropout values (voltages) (see 3.12.2). Specified pickup or latch/reset, hold, and dropout values (voltage) shall be measured as specified in 4.8.8.2.1, 4.8.8.2.2, 4.8.8.2.3, or 4.8.8.2.4. For qualification inspection, the relay shall be mounted in each of three mutually perpendicular planes, unless otherwise specified (see 3.1). For quality conformance inspection, the relay mounting position is optional. A suitable indicating device shall be used to monitor contact position. During the test, all contacts shall not change state (break or remake) when the coil is energized at or above the specified pickup or latch/reset value (voltage); when the coil voltage is lowered to any value at or above specified hold value (voltage); or when the coil voltage is reduced from the specified dropout value (voltage) to zero. It is permissible to perform specified pickup or latch/reset, hold, and dropout values (voltages) using step function voltage changes as specified in 4.8.8.2.1 or 4.8.8.2.2, 4.8.8.2.3, and 4.8.8.2.4. Due to the fact that slow ramping will overheat relay coils and vary specified pickup or latch/reset, hold, and dropout values (voltages), the step function method shall be the governing method in cases of dispute.

4.8.8.2.1 Specified pickup value (voltage) (not applicable to latching relays). The voltage shall be increased from zero until the relay operates, and the specified pickup value (voltage) shall be measured. It is permissible to perform specified pickup value (voltage) using a step function voltage change as follows and as illustrated on figure 3.

a. Step up to the maximum specified pickup value (voltage), contacts should have transferred and all normally-open contacts should be made.

b. Increase to rated coil voltage.

c. Step down to specified hold value (voltage). Normally-open contacts must still be making.

d. Step down to specified dropout value (voltage), all contacts should have transferred and all normally-closed contacts should be making.

e. Step down to zero voltage.

4.8.8.2.2 Specified latch/reset value (voltage) (applicable to latching relays only). Before measuring the specified latch/reset value (voltage), establish that all contacts are in their last energized mode for latching relays or in the plus (+) contrasting color bead mode for single coil relays (see 3.1). If not, apply rated voltage to establish last energized contact position. Gradually increase the voltage to the latching coil or in the latching direction for single coil relays until the contacts transfer and the specified operate (latch) value (voltage) shall be measured. Apply rated latching voltage and reduce to zero. Gradually increase the voltage to the reset coil or in the reset direction for single coil relays until the contacts transfer and the specified reset value (voltage) shall be measured. It is permissible to perform maximum latch/reset value (voltage) using a step function as follows and as illustrated on figure 3.

a. Step up to specified latch value (voltage) for latching relays or in the latch direction for single coil relays. The contacts shall have transferred to the latch position.

b. Step up to rated voltage and then step down to zero.

c. Step up to specified reset value (voltage) for latching relays or in the reset direction for single coil relays. The contacts shall have transferred to reset position.

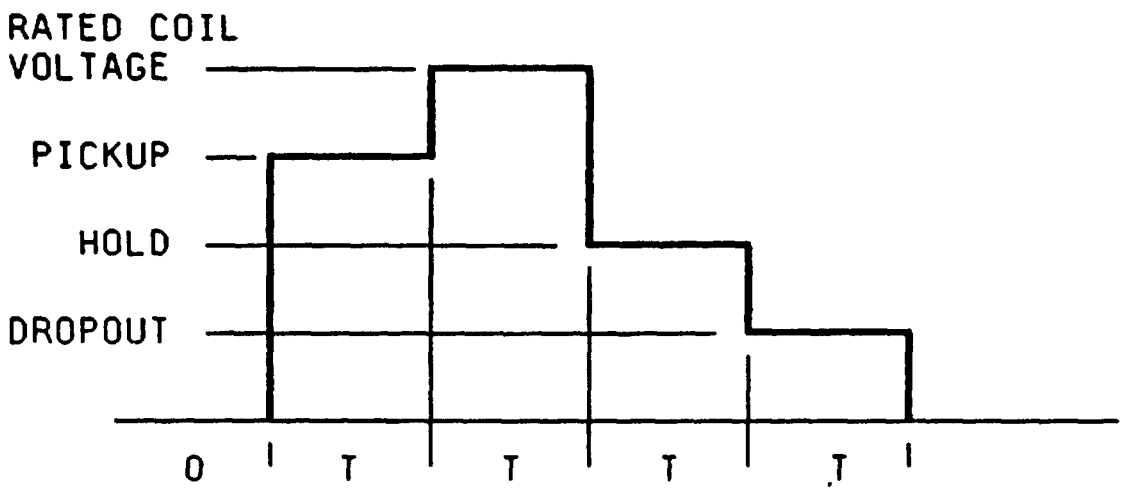
d. Step up to rated voltage and then step down to zero.

4 8.8.2.3 Specified hold value (voltage) (not applicable to latching relays). Rated voltage shall be applied to the coil. The voltage shall then be reduced to the specified hold value (voltage) (see 3.1 and 6.8). All contacts shall not change state. It is permissible to perform this test using the step function voltage program described in 4.8.8.2.1 and figure 3.

4 8.8.2.4 Specified dropout value (voltage) (not applicable to latching relays). Rated voltage shall be applied to the coil. The voltage shall then be gradually reduced until all contacts return to the de-energized position and the specified dropout value (voltage) shall be measured. It is permissible to perform this test using the step function voltage program described in 4.8.8.2.1 and figure 3.

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Nonlatching relays



Latching relays

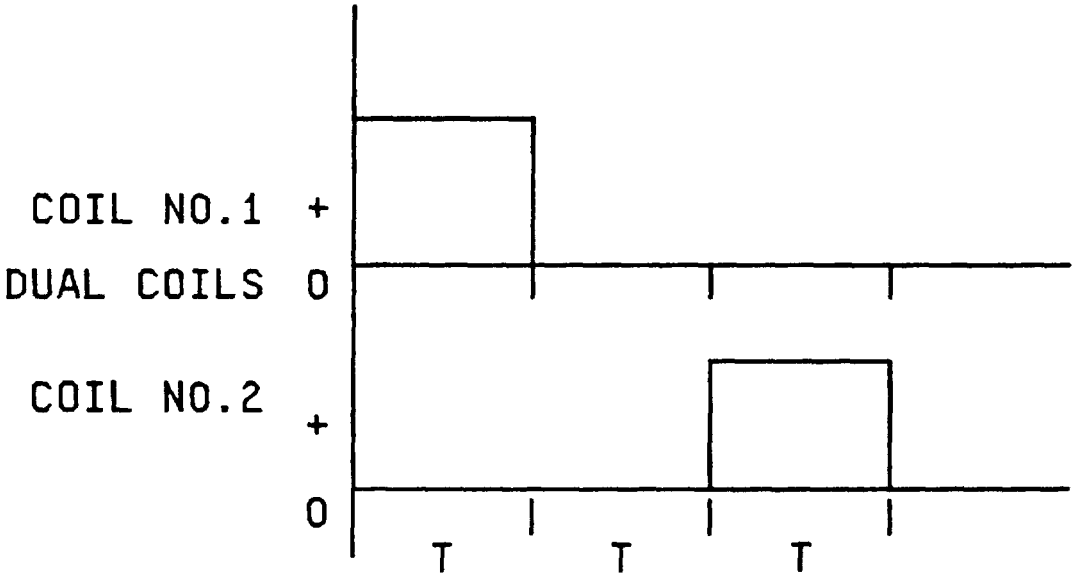


FIGURE 3 Pickup or latch/reset hold and dropout sequencing.

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4.8.8.3 Coil resistance or coil current (see 3.12.3).

4.8.8.3.1 Coil resistance (see 3.12.3.1). Relay coils shall be tested in accordance with method 303 of MIL-STD-202.

4.8.8.3.2 Coil current (when specified see 3.12.3.2). When rated voltage is applied to the coil terminals, the current must be within the limits as specified (see 3.1). Apply rated voltage for 5 seconds maximum.

4.8.8.4 Operate and release time (see 3.12.4). Operate and release time of all contacts shall be measured using an oscilloscope or other acceptable means approved by the qualifying activity. Rated voltage shall be applied to the coil. Contact load conditions shall be 6 volts dc maximum or peak ac at 10 mA maximum. The circuit shown on figure 4, or equivalent, shall be used. The operate and release time shall be exclusive of contact bounce or contact stabilization time. Timing measurements shall be made on all contact sets. Release time is not applicable to latching relays.

4.8.8.4.1 Break before make (see 3.12.4.1). The circuit shown on figure 5 or an equivalent circuit approved by the qualifying activity shall be used to monitor contact position.

4.8.8.5 Contact dynamic characteristics (see 3.12.5).

4.8.8.5.1 Contact bounce (applicable to failure level rate "L") (see 3.12.5.1). Contact bounce shall be measured on each contact set using an oscilloscope or other acceptable means approved by the qualifying activity. The trace shall show contact switching at operate and release and appropriate timing markers when using an oscilloscope. Rated voltage shall be applied to the coil. Contacts shall be loaded with 6 volts dc maximum or peak ac at 10 mA maximum. After high level rated load life and intermediate current tests, contact bounce shall be measured at 100 mA maximum at 28 V dc maximum. A contact bounce shall be considered any occurrence equal to or greater than 90 percent of the open circuit voltage with a pulse width of 10 microseconds or greater. The circuit shown on figure 4, or equivalent, shall be used.

4.8.8.5.2 Contact stabilization time (applicable to failure rate levels "M", "P", and "R") (see 3.12.5.2). Contact stabilization time shall be measured on each contact set using an oscilloscope or other acceptable means approved by the qualifying activity. The equipment shall have the capability of indicating a failure pulse width of greater than 1 µs. The trace shall show contact switching at operate and release, appropriate timing markers and maximum allowable contact voltage drop markers (which equates to the allowable static contact resistance) when measured using an oscilloscope. Rated voltage shall be applied to the coil. Contacts shall be loaded with 50 mV dc maximum or peak ac at 50 mA maximum. Contact stabilization time shall be defined as the maximum time allowed for the contacts to reach and maintain a static contact resistance state following the actual operate or release time of the relay (essentially, it is the sum of the contact bounce time and the time for the dynamic contact resistance to stabilize to the static contact resistance). After high level rated load life and intermediate current tests, the contacts shall be loaded at 100 mA maximum at 28 V dc maximum and contact bounce in lieu of contact stabilization time shall be measured as specified in 4.8.8.5.1.

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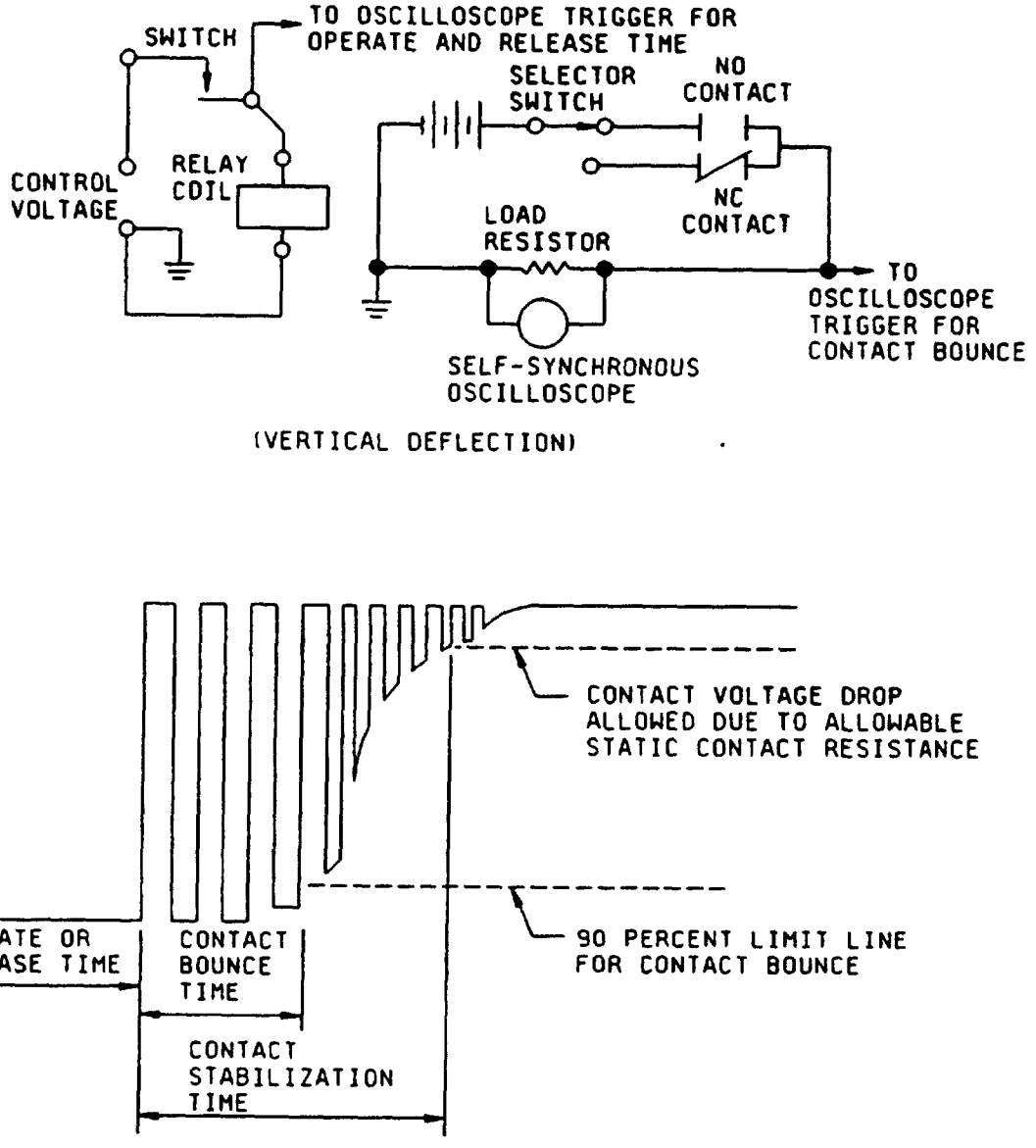
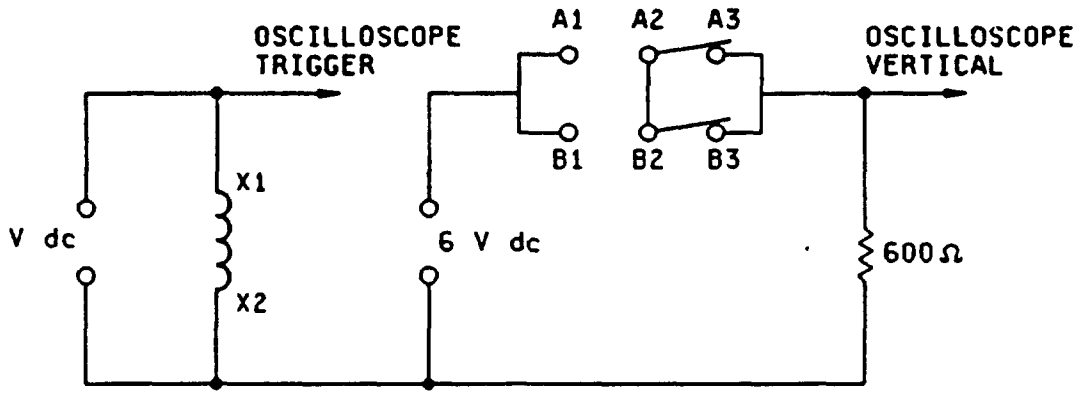


FIGURE 4. Typical circuit for operate and release time, contact bounce and contact stabilization time with typical traces.

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Nonlatching relays



Latching relays

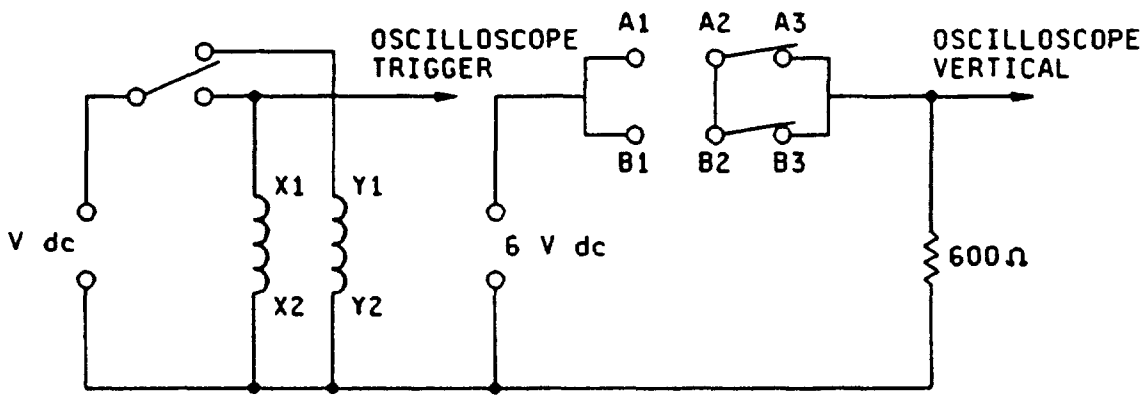


FIGURE 5. Typical break before make test circuits.

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4.8.8.6 Coil transient suppression and diode block integrity test (3.12.6).

4.8.8.6.1 Coil transient suppression (applicable to dc operated relays with internal coil transient suppression (see 3.12.6.1). The coil shall be connected as shown on figure 6 or by an equivalent test circuit.

NOTE: Voltage greater than the maximum specified may damage the coil transient suppression device.

a. The switching relay shall be a mercury wetted contact switching relay. E is the rated coil voltage for the relay under test. The switching relay is operated from a source voltage independent of E. CRO is cathode-ray oscilloscope with a rise time of .020 μs or less. The horizontal (time) deflection scale shall be set at 0.5 to 1 ms per division (.5 to 1 ms/cm), and the vertical (voltage) deflection to be such that the vertical gain provides accuracy in reading. Example: 5 V/cm for a 12 V back EMF limit, or 10 V/cm for a 24 V back EMF limit. Both horizontal (time) and vertical (voltage) deflection traces shall be of a calibrated grid spacing (i.e., cm or mm).

b. The source, E, shall be a low impedance source capable of delivering the rated coil supply voltage with no limiting resistor or potentiometer used to regulate the line voltage.

c. The switching relay shall be closed for a minimum of ten times the operate time of the test relay, or switching relay whichever is longer, to allow the oscilloscope and circuit network to stabilize and then opened to obtain the induced voltage deflection trace. The relay is to be driven at a 10 ±2 Hz cycling rate with approximately equal open and closed times.

(1) The reading shall be observed on the oscilloscope or other equivalent electronic test instrumentation. The magnitude of the induced voltage transient shall be noted. A typical trace is presented on figure 7.

(2) The maximum value of three consecutive readings shall be noted.

d. For group A testing, only one measurement and other suitable test methods may be used to verify that the back EMF (coil kick) is within the specified limit.

4.8.8.6.2 Diode block integrity test (applicable to dc operated relays with polarity reversal protection) (see 3.12.6.2). With applicable voltage (see 3.1) applied to the relay coil circuit in the reverse direction, monitor leakage current with a dc microammeter or oscilloscope or qualifying activity approved test equipment. Leakage current shall not exceed the value specified (see 3.1).

4.8.8.7 Neutral screen (applicable to latching relays only) (see 3.12.7). Relays shall be tested as follows:

a. Apply rated coil voltage to both coils simultaneously for a period of 10 ms minimum. After voltage is removed determine if the relay contacts are in a neutral position.

b. If the relay contacts will not maintain a neutral position, repeat step 4.8.8.7a. twice. A relay which will not assume a neutral position for these three successive cycles is considered an acceptable part and does not require further testing. Relays which remain in a neutral position shall tested as follows:

(1) Apply a 10 ±1 ms pulse of the maximum allowable (at +25°C) latch voltage to the latch coil. Verify that the relay has latched. Failure to latch shall be cause for rejection.

(2) Repeat step 4.8.8.7a. above.

(3) Apply a 10 ±1 ms pulse of the maximum allowable (at +25°C) reset voltage to the reset coil. Verify that the relay has reset. Failure to reset shall be cause for rejection.

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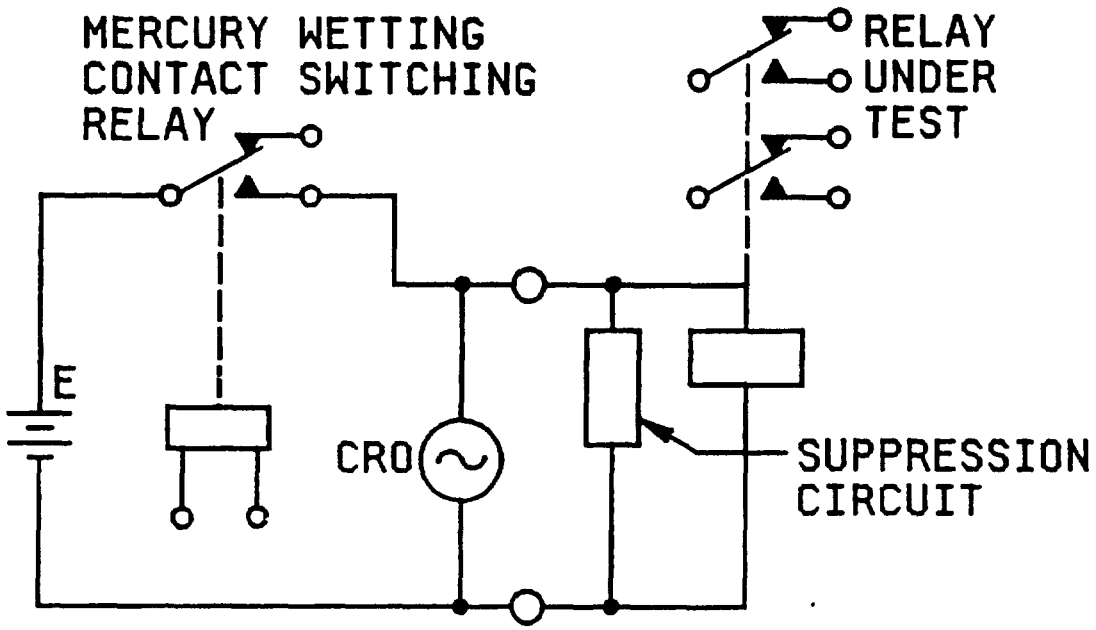


FIGURE 6. Coil transient suppression test circuit.

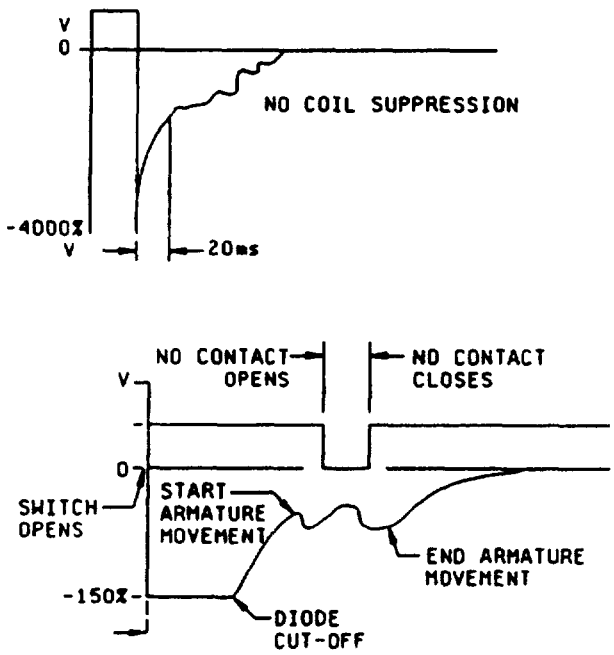


FIGURE 7. Typical transient voltage.

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4.8.9 Thermal shock (see 3.13). Relays shall be tested in accordance with method 107 of MIL-STD-202. The following details and exceptions shall apply:

a. Special mounting: Relays shall be suspended in the test chamber by twine, test socket, or other nonheat-conducting material. Test leads may be used for mounting; however, they shall not exceed the diameter or cross section of the device lead. (NOTE: When test sockets are used, the surface of the header shall not directly contact with the surface of the socket and the leads of the socket shall not exceed the diameter or cross section of the device lead).

b. Test condition: B, except exposure time at temperature extreme during the fifth cycle shall be for 2 hours each.

c. Measurements at each temperature extreme during steps 1 and 3 of this fifth cycle at the end of each temperature exposure, and with the relays still in the conditioning chamber, the insulation resistance, specified pickup or latch/reset and dropout values (voltages) and operate and release time, shall be measured as specified in 4.8.6, 4.8.8.2, and 4.8.8.4, respectively. Specified pickup or latch/reset and dropout values (voltages) shall be measured in any position.

d. Examination after test: Relays shall be visually examined for cracking, peeling, and flaking of the finish, and the dielectric withstanding voltage shall then be measured as specified in 4.8.7.1.

4.8.10 Shock (specified pulse) (see 3.14). Unless otherwise specified (see 3.1), relays shall be tested in accordance with method 213 of MIL-STD-202. The following details and exceptions shall apply:

a. Mounting method: For relays having flanges or studs, mounting shall be by the flanges or studs; and for relays without flanges, mounting shall be by plugging into sockets, or other suitable means approved by the qualifying activity.

b. Test condition: A, B, or C, as applicable (see 3.1).

c. Electrical-load conditions: In each direction of shock, the coil shall be de-energized during two shocks and energized with rated voltage during one shock.

d. For latching relays: In each direction there shall be two pulses in the latch position and two pulses in the reset position, with no coil voltage being applied during these pulses.

e. Measurements during shock: Contacts shall be monitored with an adequate test circuit as approved by the qualifying activity. The test circuit shall verify that no opening of closed contacts in excess of 10 μs and no false closure or bridging of open contacts occurs in excess of 1 μs. The contact load shall be 10 mA maximum at 6 V dc maximum (dc or peak ac).

f. Examination after test: Relays shall be examined for evidence of structural failure or other damage which might impair the operation of the relay.

4.8.11 Vibration (see 3.15).

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4.8.11.1 Vibration (sinusoidal). Unless otherwise specified (see 3.1), relays shall be tested in accordance with method 204 of MIL-STD-202. The following details and exceptions shall apply:

a. Method of mounting: As specified in 4.8.10a.

b. Electrical-load conditions: For qualification and group C inspections, relays shall be tested with the coil energized for 2 hours at rated voltage and with the coil de-energized for 2 hours, in each of the three mutually perpendicular (x, y, and z) directions (For latching relays, relays shall be tested with the coils de-energized; 2 hours with the latch coil last energized and 2 hours with the reset coil last energized, in each of the 3 mutually perpendicular relay axes). Contacts shall be loaded as specified in 4.8.11.1d.

c. Test condition: D, except vibration level is the lesser of .195 inch (4 95 mm) double amplitude or 30 G, and frequency range is 10 to 3,000 Hz.

d. Tests during vibration: As specified in 4.8.10d.

e. For group A testing, only one cycle shall be performed over the frequency range of 100 Hz to 3,000 Hz unless otherwise specified (see 3.1). The relay shall be vibrated in the direction of contact motion. At the option of the manufacturer, the relay production lot may be divided in three equal groups and each group shall be tested in one of three mutually perpendicular axes (X, Y, and Z). When relays are tested as specified in the optional procedure, a failure of any relay in any axis shall require 100 percent inspection of all relays in the production lot in each axis. For all contact forms except contact form "A", the cycle shall consist of 3 minutes up with the coil energized with rated voltage and 3 minutes down with coil de-energized. The cycle for form contact "A" shall consist of 3 minutes up with the coil de-energized and 3 minutes down with the coil energized. Latching relays shall remain in each latched position with no voltage applied to the coils 3 minutes in the latch position and 3 minutes in the reset position.

f. Examination after test: Prior to removal from test fixture and without disturbing the relay, apply maximum over the temperature range pickup voltage value (see 3.1) to the coil, and then verify that relay contacts transfer. Remove coil voltage, and then verify that relay contacts transfer. For latching relays, apply the maximum over the temperature range latch voltage and verify that the relay contacts transfer, then apply the maximum over the temperature reset voltage and verify that the relay contacts transfer. Failure of relay contacts to transfer shall be cause for rejection. After removal from test fixture, the relays shall then be inspected for evidence of structural failure or other damage which might impair the operation of the relay.

4.8.11.2 Vibration (random). Relays shall be tested in accordance with method 214 of MIL-STD-202. The following details and exceptions shall apply:

a. Method of mounting: As specified in 4 8.10a.

b. Electrical-load conditions: For qualification and group C inspections, relays shall be tested with the coil energized at rated voltage for 15 minutes and with the coil de-energized for 15 minutes, in each of three mutually perpendicular (x, y, and z) directions. Latching relays shall be tested with the coils de-energized; 15 minutes with the latch coil last energized and 15 minutes with the reset coil last energized. When applicable to group A inspection, the test time shall be 3 minutes for each state described above. Contacts shall be monitored as specified in 4.8.11.2d.

c. Test condition: IG (0.4 G^2/Hz, 23.91 rms G).

d. Tests during vibration: As specified in 4.8.10d.

e. Examination after test: As specified in 4 8.11.1f

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4.8.12 Acceleration (unless otherwise specified, see 3.1 and 3.16). Relays shall be tested in accordance with method 212 of MIL-STD-202. The following details and exceptions shall apply:

a. Mounting of specimens: As specified in 4.8.10a.

b. Test conditions: Test condition A, 50 G's, unless otherwise specified (see 3.1) (acceleration shall be applied in each direction along three mutually perpendicular axes (x, y, and z) of the specimen; one axis shall be in the direction most likely to cause malfunction). In each direction, the coil shall be de-energized for 5 minutes, rated coil voltage shall be momentarily applied; and the voltage shall be reduced to the maximum ambient pickup voltage as specified (see 3.1) for 5 minutes. Latching relays shall remain in each latched position with no voltage applied to the coils. Contacts shall be monitored for proper position.

c. Examination after test: As specified in 4.8.10f.

4.8.13 Terminal strength (see 3.17). Relays shall be tested in accordance with method 211 of MIL-STD-202 in accordance with the following, as applicable. Unless otherwise specified herein, two terminals of each discrete design, size, and configuration shall be tested; however, if there is only one of such design, size, and configuration, it shall be tested.

4.8.13.1 Pull test (all terminal types). Terminals shall be tested as specified in test condition A, the force shall be as specified (see 3.1).

4.8.13.2 Bend test (all terminal types). Terminals shall be tested as specified in test condition B (two bends) or C, as applicable. Loads for test condition C shall be as specified (see 3.1).

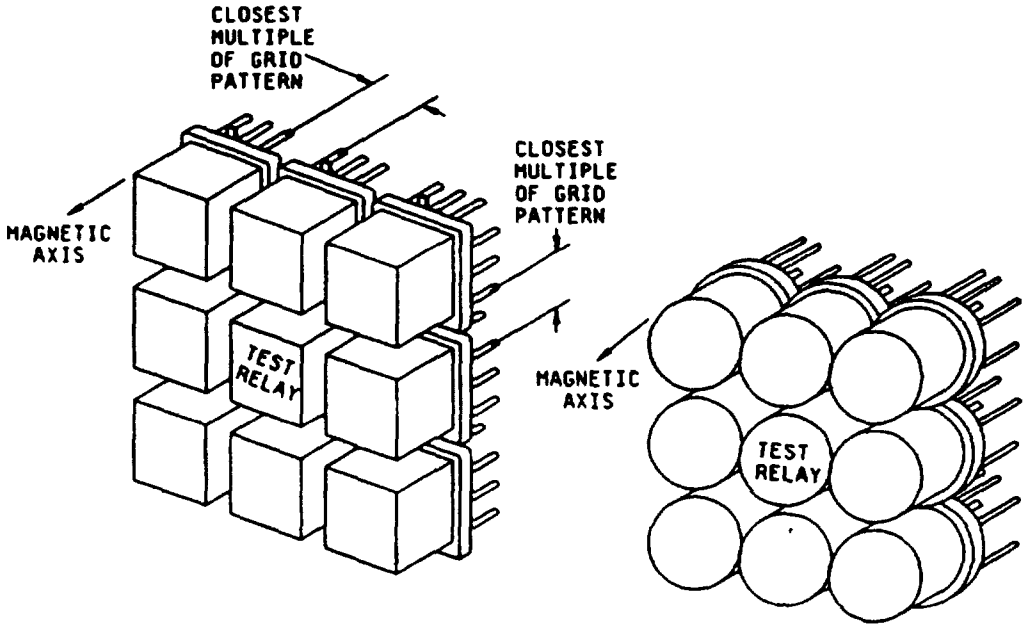
4.8.13.3 Twist test (wire lead terminals only) (not applicable to relays with spreader pads or leads less than .50 inch (12.7 mm) in length). All terminals shall be tested as specified in test condition D, except during application of torsion, each terminal shall be rotated 45° in one direction, then returned to start; rotated in opposite direction 45°, then returned to start. Each terminal shall be subjected to two such rotations and returns. Each terminal shall be held at a point .75 inch (19.0 mm) from the point of emergence from the relay, or terminal end for relays with terminals less than .75 inch (19.0 mm), and in one plane shall be bent 20° ±5° in one direction, then returned to start; bent in opposite direction 20° ±5°, then returned to start. This procedure shall then be repeated in the perpendicular plane. Following these tests, relays shall be examined for evidence of loosening or breaking of the terminals and other damage that could adversely affect the normal operation of the relay; the requirements of 3.30 are not applicable.

4.8.14 Magnetic interference (when specified, see 3.1 and 3.18). The relay under test and eight similar relays shall be mounted in the same physical orientation by nonmagnetic means, as shown on figure 8. Unless otherwise specified, grid-spaced relays shall be mounted so that all terminals are positioned in the closest possible grid pattern. The spacing between terminals and tops of adjacent terminals shall be as shown on figure 8 unless otherwise specified. Pickup or latch/reset and dropout voltage shall be measured on the relay under test as specified in 4.8.8.2 with the coils of the eight outer relays energized at rated voltage (the magnetic polarity of each relay shall be similarly oriented). This measurement shall be repeated with the coils of the eight outer relays de-energized.

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0.100 grid pin pattern

TO-5, round pin pattern



Non-TO-5 patterns

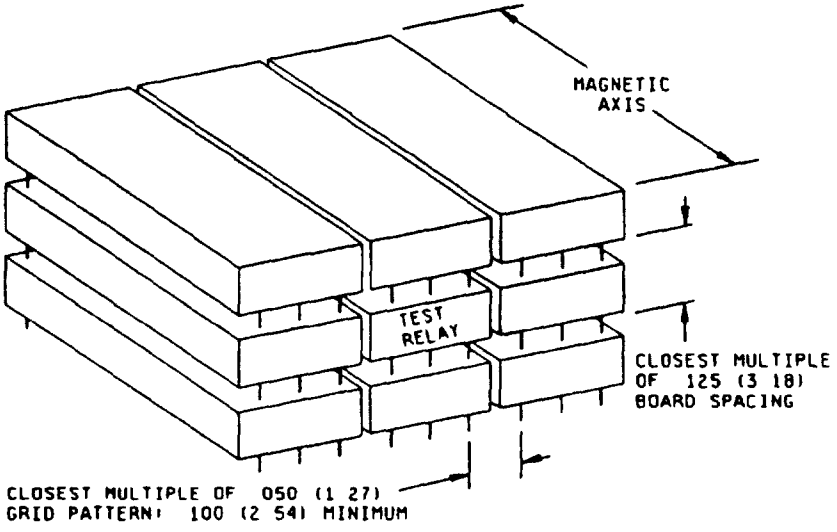


FIGURE 8. Mounting arrays for adjacent similar relays.

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4.8.15 Coil life (unless otherwise specified, see 3.1) (see 3.19). Relays shall be tested for 1,000 hours as follows (see figure 9). For latching relays, the test time shall be equally divided between the two coils.

a. Relays may be mounted as specified in 4.3.5. Each contact terminal shall be connected as specified in 4.8.8.1a.

b. During the maximum temperature portion of the test, rated coil voltage (see 3.1) shall be applied continuously and at least half of the normally open contacts shall carry rated current. During room temperature and minimum temperature exposures, the coil shall be de-energized and contacts shall not be loaded. Ambient temperatures shall be varied as shown on figure 9, with heating and cooling rates not to exceed +1°C per second average. The portion of the cycle run at minimum temperature shall be approximately 10 percent of the test cycle time.

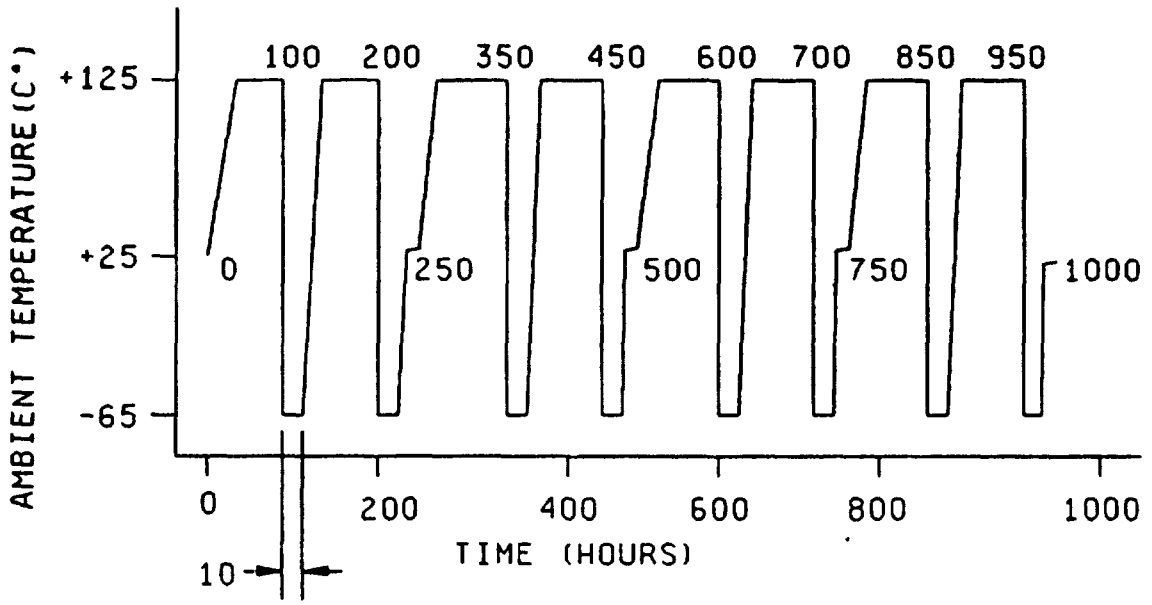
c. After the first 100 hours, and while still at the low temperature extreme, contact resistance (or voltage drop) and operate and release time shall be measured as specified in 4.8.8.1 and 4.8.8.4 and the values shall be recorded. Measurements of coil resistance (or coil current as specified, see 3.1) and contact resistance as specified in 4.8.8.3 and 4.8.8.1, shall be taken at room temperature initially, then again after 250 ±25 hours, 500 ±25 hours, and 750 ±25 hours and the values shall be recorded. Specified pickup, hold, and dropout values (voltages) measurements shall be made as specified in 4.8.8.2 during the last temperature cycle after allowing the relay to stabilize, de-energized, at each of the temperature extremes, and the values shall be recorded. For latching relays, after the first 50 hours, and while still at the low temperature extreme, contact resistance and operate time shall be measured as specified in 4.8.8.1 and 4.8.8.4 respectively and the values shall be recorded. Measurements of coil resistance or coil current as applicable (see 3.1), and contact resistance as specified in 4.8.8.1 shall be taken at room temperature initially, then again after 125 hours ±12.5 hours, 250 hours ±12.5 hours, and 375 hours ±12.5 hours and the values shall be recorded. Specified latch and reset voltage measurements shall be made as specified in 4.8.8.2.2 during the last temperature cycle after allowing the relays to stabilize, de-energized, at each of the temperature extremes, and the values shall be recorded.

d. After the relay(s) have been subjected to the coil life temperature cycles, measurements shall then be taken at room temperature of dielectric withstanding voltage, insulation resistance, contact resistance, coil resistance (or coil current), operate and release time, and contact bounce as specified in 4.8.7.1, 4.8.6, 4.8.8.1, 4.8.8.3, 4.8.8.4, and 4.8.8.5, respectively.

e. Examination after test: As specified in 4.8.10f.

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Nonlatching relays



Latching relays (per coil)

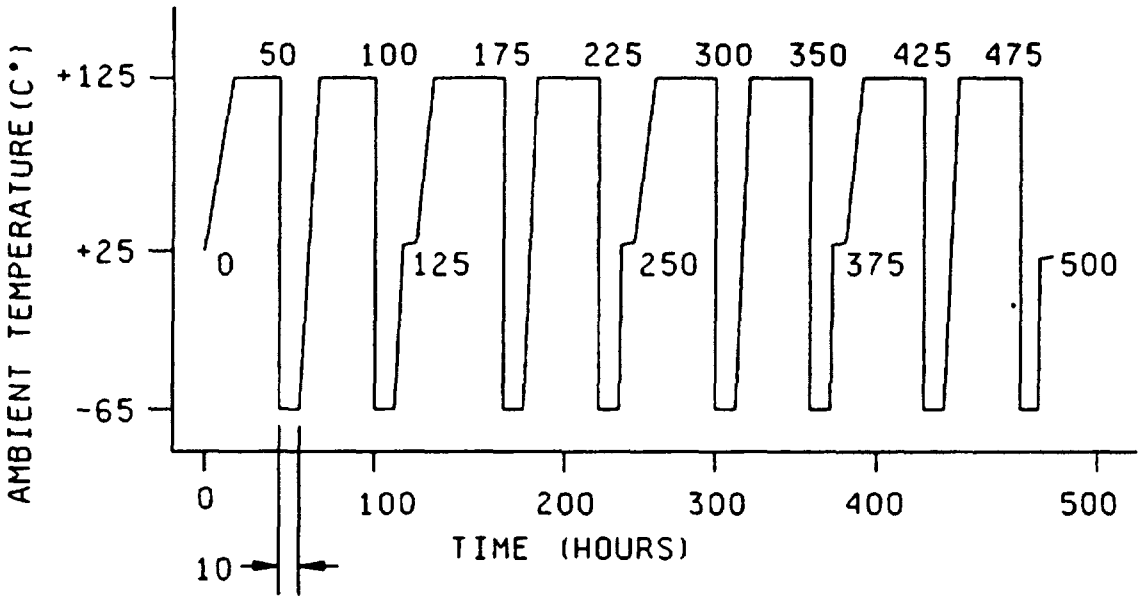


FIGURE 9 Coil life ambient temperatures

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4.8.15.1 Coil endurance (see 3.19.1). Relays shall be tested for 114 hours minimum as follows:

a. The coil resistance shall be tested in accordance with 3.12.3 prior to testing and shall be as specified (see 3.1).

b. Relays may be mounted as specified in 4.3.5. Each contact terminal shall be connected as specified in 4.8.8.1a.

c. The test chamber shall be controlled to maintain the maximum specified ambient temperature (see 3.1).

d. Apply the maximum coil voltage (see 3.1) at a cycling rate of 4 ±.4 seconds on and 2 ±.2 seconds off for 114 hours minimum.

e. For latching relays, the test time shall be equally divided between the two coils.

f. Following the test, the coil resistance shall be measured in accordance with 3.12.3 and shall be as specified (see 3.1).

4.8.16 Resistance to soldering heat (see 3.20) Relays shall be tested in accordance with method 210 of MIL-STD-202. The following details and exceptions shall apply:

a. Depth of immersion in molten solder: Within .060 ±.020 inch (1.52 ±0.51 mm) of the relay base.

b. Test condition: B.

c. Measurements after test: Insulation resistance, contact resistance, specified pickup or latch/reset, hold, and dropout values (voltages), and coil resistance (or coil current as specified, see 3.1) shall be measured as specified in 4.8.6, 4.8.8.1, 4.8.8.2, and 4.8.8.3, respectively.

d. Examination after test: As specified in 4.8.10f.

4.8.17 Salt spray or salt atmosphere (corrosion) (see 3.21). Relays shall be tested as specified in 4.8.17.1 or, when specified (see 3.1), in accordance with 4.8.17.2.

4.8.17.1 Salt spray, MIL-STD-202 (see 3.21). Relays shall be tested in accordance with method 101 of MIL-STD-202 unless otherwise specified (see 3.1). The following details and exceptions shall apply:

a. Applicable salt solution: 5 percent.

b. Test condition: B.

c. Examination after test: Relays shall be examined for evidence of peeling, chipping, blistering of the finish, and exposure of base metal due to corrosion that can adversely affect the operation or performance of the relay.

4.8.17.2 Salt atmosphere (corrosion), MIL-STD-750 (when specified, see 3.1). Relays shall be tested in accordance with method 1041 of MIL-STD-750. The following detail shall apply: Examination after test: As specified in 4.8.17.1c.

4.8.18 Overload (applicable to high level relays only) (see 3.22). Relay contacts (both normally open and normally closed) shall be subjected to the tests specified in 4.8.18.1 through 4.8.18.2, as applicable. For double-throw relays, separate tests shall be performed for the normally open and the normally closed contacts. Relay coil energization shall be nonsynchronous with the power supply for ac loads. The number of cycles shall be 100 for dc contact loads, and 200 for ac contact loads ("on" and "off" times shall be approximately equal). The coil shall be energized at rated voltage. Overload current shall be twice rated load current. The opening and closing of all contacts under test shall be continuously monitored to detect switching of the contacts. The equipment shall be capable of monitoring 40 percent minimum of the actual opened and closed time of each contact. The equipment shall automatically cut off when a failure occurs, or shall record any failures The relay case shall be connected to system ground through a normal-blow fuse rated at 5 percent of the test current or 100 mA, whichever is greater. After the test, the fuse shall be tested for electrical continuity.

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4.8.18.1 Resistive load. Suitable noninductive resistors shall be used. Cycling rate shall be 20 ±2 cycles per minute (cpm).

4.8.18.2 Inductive load. Cycling rate shall be 10 ±1 cpm. Inductive load testing shall be performed as recommended by MIL-HDBK-338 and approved by the qualifying activity.

4.8.18.2.1 Inductive load, dc. Unless otherwise specified (see 3.1), dc inductive loads shall be computed in accordance with the following procedure:

a. The relay shall be mounted in its normal operating position on a nonmagnetic plate with no immediately adjacent metal.

b. The load circuit shall be energized through a bounce free contact from a power supply regulated to within 5 percent of the specified rated voltage as shown on figure 10.

c. By measuring the time for the current to reach 63 percent of its final value, the dc inductance is calculated as:

$L = Rt$ where

L is the inductance in henries,
R is the resistance of the load circuit, and
t is the time in seconds for current to reach 63 percent of its final value.

4.8.18.2.2 Inductive load, ac. The load shall consist of inductive and resistive elements with 0.7 ±0.05 lagging power factor at the voltage and frequency specified (see 3.1).

4.8.19 Life (see 3.23 and 6.8). Unless otherwise specified (see 3.1), relays shall be operated for 100,000 cycles with contacts loaded in accordance with 4.3.3 and 4.8.19.1 through 4.8.19.4, as applicable. Relay coil energization shall be nonsynchronous with the power supply for ac loads. Each relay case shall be connected to system ground through an individual normal-blow fuse rated at the greater of 100 mA or 5 percent of load current. The ambient temperature of the interior of the test chamber shall be +125°C minimum. The contact resistance or voltage drop, as applicable, shall be monitored during 40 percent minimum of each "on" and each "off" period. The monitoring equipment shall automatically hold the relay under test in the state in which the failure occurs (energized or de-energized) and the contact voltage shall be removed. All contacts in each sample unit shall switch identical loads. Following cycling, the electrical continuity of each case-grounding fuse shall be checked.

4.8.19.1 Resistive load. Cycling rate shall be 20 ±2 cycles per minute. Suitable resistors shall be used. Current shall be rated resistive current as specified (see 3.1).

4.8.19.2 Inductive load. Cycling rate shall be 10 ±1 cycles per minute with approximately equal "on" and "off" periods. Current shall be rated current. Appropriate inductive load components (see 4.8.18.2.1 and 4.8.18.2.2, as applicable) shall be used. A suitable resistor may be placed in the circuit to obtain rated steady-state current flow. Inductive load testing shall be performed as recommended by MIL-HDBK-338 and approved by the qualifying activity.

4.8.19.3 Lamp load. Unless otherwise specified, the lamp load shall be performed with a 28 V dc power supply voltage. Either lamps or simulated loads may be used to provide the load as follows:

a. Simulated loads shall subject the relays to making 12 times the rated lamp load and breaking the rated lamp load. The duration of the 12 times inrush shall be 0.015 to 0.020 second. The total "ON" time shall be 5 ±0.05 seconds and the "OFF" time shall be 5 ±0.05 seconds.

b. Lamps shall be screened to verify a minimum initial cold inrush of 12 times the rated lamp load. Only lamps meeting this requirement shall be used to provide load to the relays. The total "ON" time shall be 2 ±0.05 seconds and the "OFF" time shall be 7 ±2 seconds.

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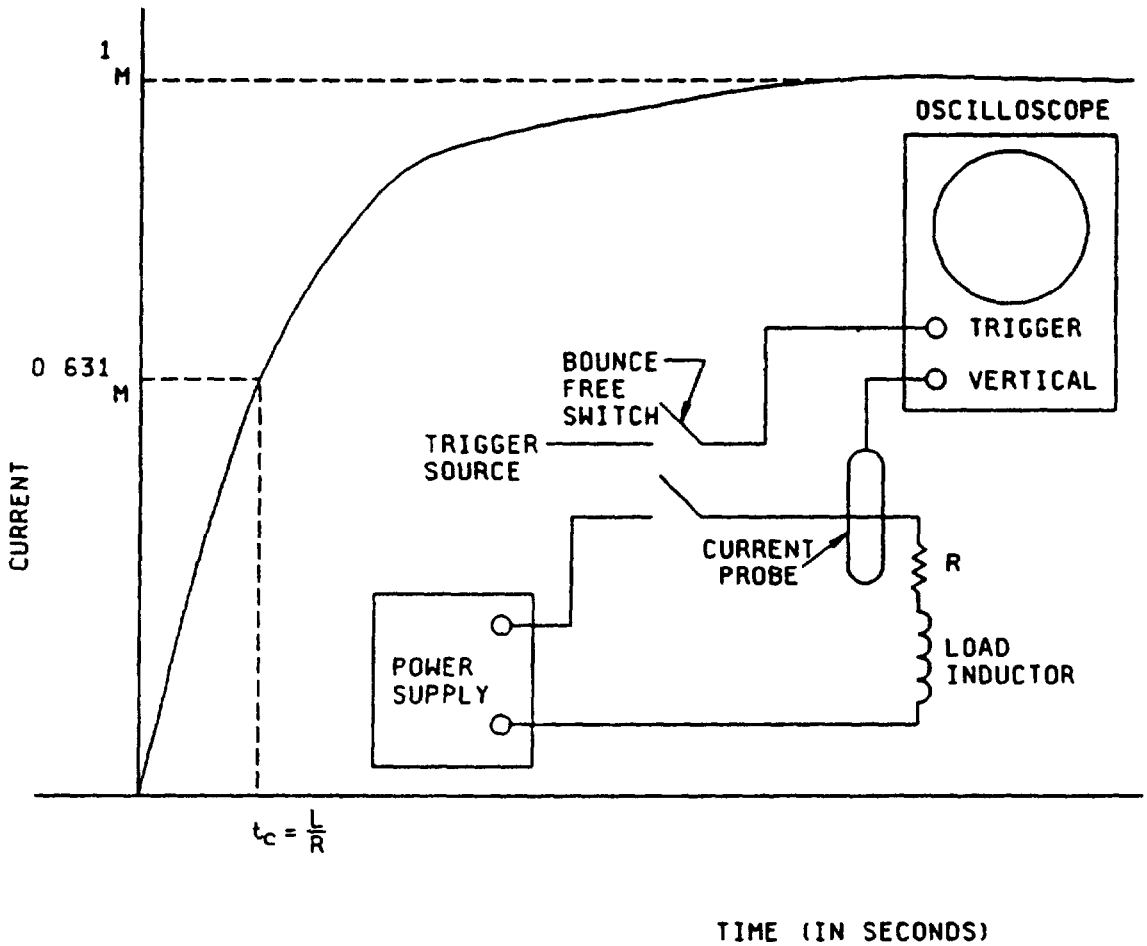


FIGURE 10. DC inductive load (inductive current rise time curve).

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4.8.19.4 Low level load. The minimum cycle time shall be 10X the sum of the maximum specified (see 3.1) operate and release times for the relay under test. For latching relays, the minimum cycle time shall be 10(2X operate time). Each contact load shall be 10 to 50 μA at 10 to 50 mV (dc or peak ac). The coil shall be energized at rated voltage during 50 ±10 percent of each operation. Example: 10 x 8 ms = 80 ms which would be a maximum of 12.5 cycles (one opening and one closure of a set of contacts) per second.

4.8.20 Intermediate current (see 3.24). Intermediate current shall not be considered a low level or high level contact load rating (see 6.1.1). Relays shall be tested as specified in method 312 of MIL-STD-202. The following details and exceptions shall apply:

a. Maximum contact resistance: As specified in 3.24.

b. Coil energizing voltage: Rated voltage (see 3.1).

c. Cycling rate: 10 ±2 cycles per minute with minimum of 75 percent coil "ON" time. For latching relays, the relay shall be in each position for approximately one-half the cycle time. Each coil shall be energized for 75 percent of each cycle.

d. Contact current/voltage: 100 mA at 28 V dc.

e. Monitoring: At least 40 percent of the closed time of each contact during each operation shall be monitored for contact resistance.

f. Number of cycles: 50,000 cycles.

g. Ambient temperature: +125°C minimum.

4.8.21 Mechanical life (see 3.25). Relays shall be cycled one million cycles at room ambient temperature. The cycling rate shall be as specified in 4.8.19.4. Each contact load shall be 10 to 50 μA at 10 to 50 mV (dc or peak ac). The coil shall be energized at rated voltage during 50 ±10 percent of each cycle. Mechanical life cycles shall not be used for failure rate determination or maintenance.

4.8.22 Resistance to solvents (see 3.26). Relays shall be tested in accordance with method 215 of MIL-STD-202 except temperature of solvents a, b, and d shall be maintained at +63°C to +70°C. The following details and exceptions shall apply:

a. Portion to be brushed: All marking.

b. Optional procedure for solvent d is not applicable.

c. Specimens to be tested: One sample shall be tested with each solvent solution. For qualification inspection, samples in excess of the available number of solvents shall be exempt from test. For group C testing, two additional samples shall be selected. The additional samples may be electrical rejects from group A2.

d. Examination: Specimens shall be examined for legibility of marking.

4.8.23 Particle impact noise detection (PIND, when specified see 3.1 and 3.27). When specified, relays shall be tested as specified in appendix B.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C, as specified (see 6.2d).

5.1.1 Level A.

5.1.1.1 Cleaning. Relays shall be cleaned in accordance with MIL-P-116, process C-1 if necessary.

5.1.1.2 Drying. Relays shall be dried in accordance with MIL-P-116 if necessary.

5.1.1.3 Preservative application. Contact preservatives shall not be used.

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5.1.1.4 Unit packs. Each relay shall be unit packed one each and protected in accordance with method III of MIL-P-116.

5.1.1.5 Intermediate containers. Intermediate container use shall be in accordance with MIL-STD-2073-1. The number of unit packs per intermediate container shall be determined by appendix B of MIL-STD-2073-1.

5.1.2 Level C. The level C preservation of relays shall conform to the MIL-STD-2073-1 requirements.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Packing and unitization or palletization. Packing and unitization or palletization shall conform to the requirements of MIL-STD-2073-1 for the level cited. When the quantity of a stock numbered item being shipped level A to a single destination comprises a shipping container whose volume is less than 1 cubic foot (.02832 cubic meter) and weight is less than 25 pounds (11.3398 kilograms), packing shall be as specified for level B.

5.3 Marking. In addition to any special or other identification marking required by the contract (see 6.2e), each unit, intermediate and exterior container shall be marked in accordance with MIL-STD-129. The complete military or contractor's type or PIN, as applicable (including the CAGE), shall be marked on all unit and supplementary packs in accordance with the identification marking provisions of MIL-STD-129.

5.3.1 Civil agencies. When specified in the contract or order (see 6.2e), the marking of domestic shipments for civil agencies shall be in accordance with FED-STD-123.

5.3.2 Electrostatic discharge sensitive (ESDS) electronic devices. Unit intermediate, and exterior packs containing ESDS devices susceptible to damage from electric field forces (see MIL-HDBK-263 for definition of electric field) shall be marked in accordance with MIL-STD-129 requirements.

5.3.3 First article and quality conformance inspections. First article and quality conformance inspections and tests shall be required as specified in MIL-P-116.

5.4 Functional requirements.

5.4.1 Rough handling test (when specified, see 6.2). When packs have been tested in accordance with MIL-P-116, all material and components comprising each pack shall be free from damage or evidence of displacement which might affect the use of the preservation method or pack. The semiconductor devices and associated accessories within the tested packs shall show no visible signs of damage. When specified in the contract (see 6 2), functional tests in accordance with the group A inspection requirements of the commodity specification shall be conducted on those semiconductor devices subjected to the rough handling test to determine freedom from operational malfunction. The examination of the devices tested under this group A inspection shall be in accordance with the visual and mechanical inspection requirements specified in method 2071 of MIL-STD-750 or the commodity specification.

5.4.2 Leakage test (when applicable) When a barrier enclosed unit pack has been tested in accordance with MIL-P-116, there shall be no evidence of moisture within the unit pack.

5.4.3 Workmanship. The quality of workmanship shall assure acceptance of the completed preservation, packing, and marking requirements in accordance with the inspections specified in MIL-P-116.

5.5 Quality assurance provisions. Quality assurance and inspection requirements shall be in accordance with MIL-P-116.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. Relays conforming to this specification are intended for use in electronic and communication equipment. Their principal areas of application are for aircraft, missiles, spacecraft and ground-support electronic and communication equipment. This does not preclude the use of these relays in other military applications. Consult MIL-STD-1346 for selection and application information.

CAUTION: The use of any coil voltage less than the rated coil voltage will compromise the operation of the relay.

CAUTION: Transfer relay contacts shall not be used to ground load in one position and to 115-volt terminal in other position. Transfer relay contacts shall not be used to transfer load between phases, nor between unsynchronized ac sources.

CAUTION: When latching relays are installed in equipment, the latch and reset coils should not be pulsed simultaneously. Coils should not be pulsed with less than the rated coil voltage and the pulse width should be a minimum of three times the specified operate time of the relay. If these conditions are not followed, it is possible for the relay to be in the magnetically neutral position.

6.1.1 Contact rating. The contact ratings of relays covered by this specification are based on load endurance tests which establish the relay capability to switch rated loads. For the purposes of this specification, contact load ratings are defined as follows:

a. Low level: 10 μA at 50 mV to 10 mA at 6 V.

b. Intermediate: 10 mA at 6 V to 100 mA at 28 V.

c. High level: Above 100 mA at 28 V.

6.1.2 Failure rate level. Failure rate level (percent per 10,000 cycles) as specified in table I, is based on an average of 10 cycles per hour, per application, for 1,000 hours in accordance with MIL-STD-690.

6.1.3 Packaging provisions. The preservation, packing, and marking specified herein are intended for direct shipments to the Government. However, at the option of the contractor or when so specified, the packaging provisions herein are also applicable for the preparation of relays for shipment from the parts contractor to the original equipment manufacturer.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Title, number, and date of the applicable specification sheet, and the PIN.

d. Levels of preservation and packing required (see 5.1 and 5.2).

e. Special or additional identification marking requirements (see 5.3).

f. Marking requirements for failure rate level (see MIL-STD-690).

6.2.1 Manufacturer provided services. The manufacturer may provide the following optional value added services when specified in the purchase order:

a. Hot solder dipping of terminals (see 3.5.7).

b. Shortening of wire leads (see 3.5.7.2).

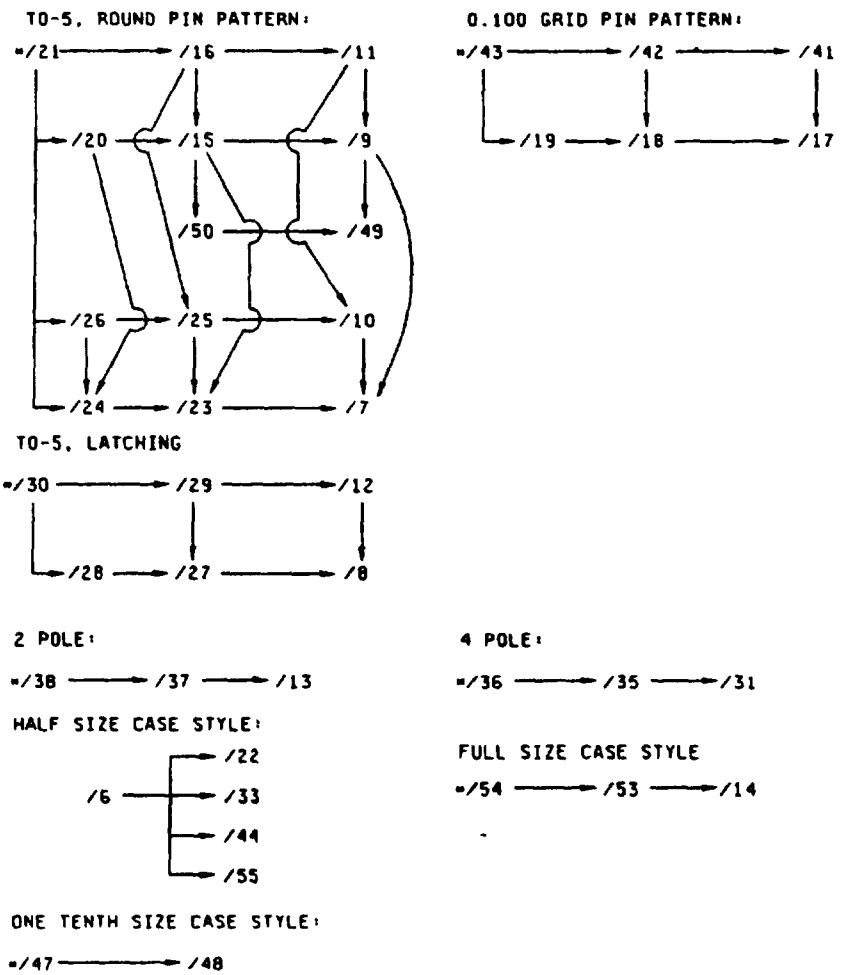
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6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in applicable Qualified Products List, QPL 39016, whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the QPL is the Defense Electronics Supply Center, ATTN: DESC-ELS, 1507 Wilmington Pike, Dayton, OH 45444. Application for qualification tests shall be made in accordance with provisions governing qualification SD-6 (see 6.3.1).

6.3.1 Provisions Governing Qualification SD-6. Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.

6.3.2 Qualification and maintenance by similarity. This information is for reference only. Refer to the individual specification sheets for detailed information.

6.3.2.1 Qualification by similarity. The graphic guide shown illustrates the relative complexity of individual specification sheets within a style group for use in determining qualification inspection (reduced testing) candidate specification sheets. Arrow direction indicates flow towards less complex designs. Asterisk indicates most complex design of style group.



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6.3.2.2 Maintenance by similarity. The data accumulated for the style groups listed below may be combined for group B and C inspections maintenance and extension of failure rate levels in accordance with MIL-STD-690. Testing samples must be representative of production. Maintenance plan is contingent upon qualifying activity approval. Failures exceeding the sampling plan selected (C = number), regardless of the specific specification sheet affected, affect all other specification sheets within the respective style group.

a. TO-5, round pin pattern: /7, /9, /10, /11, /15, /16, /20, /21, /23, /24, /25, /26.

b. 0.100 grid pin pattern: /17, /18, /19, /41, /42, /43.

c. TO-5, latching: /8, /12, /27, /28, /29, /30.

d. Half size case style: /6, /22, /33, /44, /55.

e. Full size case style: /14, /53, /54.

f. 2 pole: /13, /37, /38.

g. 4 pole: /31, /35, /36.

6.4 Shelf life. Relays conforming to this specification should be designed for a minimum of 7 years shelf life, following which the relays should be satisfactory for intended use without further conditioning.

6.5 Sealing. The following procedure is suggested as a guide; however, every effort should be made to utilize the most effective procedure consistent with the state of the art.

a. Evacuate to less than 200 microns.

b. Heat to maximum rated ambient temperature with continued evacuation (see 3.1).

c. Maintain heat and vacuum for 12 hours or longer, continuing the treatment until a maximum pressure of 80 microns is reached.

d. Turn off heaters and maintain pressure for 4 hours.

e. Close evacuation valve and fill chamber with the desired pressurizing gas.

f. Seal relay before removing from chamber.

6.6 Intermetallic contact. The finishing of metallic areas to be placed in intimate contact by assembly presents a special problem, since intermetallic contact of dissimilar metals results in electrolytic couples which promote corrosion through galvanic action. To provide the required corrosion protection, intermetallic couples are restricted to those permitted by table IX. Table IX shows metals and alloys (or plates) by groups which have common electromotive forces (EMF) within 0.05 volt when coupled with a saturated calomel electrode in sea-water at room ambient temperatures. All members of a group are considered as completely compatible, one with the other. Compatible couples between groups have been specified in table IX based on a potential difference of 0.25 volt maximum. To simplify any arithmetic involved, table IX shows, in addition to EMF against a calomel electrode, a derived "anodic index" with group 1 (gold, etc.) as 0 and group 18 (magnesium, etc.) as 175. Subtraction of a lower group anodic index gives the EMF difference in hundredths of a volt.

6.6.1 Groups. Table IX sets up 18 primary groups. It may be noted that neither the metallurgical similarity or dissimilarity of metals is the parameter for selection of compatible couples. All members within a group, regardless of metallurgical similarity, are considered inherently nonsusceptible to galvanic action, when coupled with any member within the group; for example, such dissimilar metals as platinum and gold. Similarly, such basically dissimilar alloys as austenitic stainless steel, silver-solder, and low brass (all members of group 5) are inherently nonsusceptible when coupled together.

6.6.2 Compatibility graphs. Permissible couple series are shown in table IX by the graphs at the right. Members of groups connected by lines will form permissible couples. A o indicates the most cathodic member of each series, a • an anodic member, and the arrow indicates the anodic direction.

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TABLE IX. Compatible couples (see 6.6). 1/

Group no.	Metallurgical category	EMF (volt)	Anodic index (0.01 V)	Compatible couples
1	Gold, solid and plated; gold-platinum alloys; wrought platinum (most cathodic)	+0.15	0	
2	Rhodium plated on silver-plated copper	+0.05	10	
3	Silver, solid or plated; high silver alloys	0	15	
4	Nickel, solid or plated; monel metal, high nickel-copper alloys	-0.15	30	
5	Copper, solid or plated; low brasses or bronzes; silver solder; German silver; high copper-nickel alloys; nickel-chromium alloys; austenitic corrosion-resistant steels	-0.20	35	
6	Commercial yellow brasses and bronzes	-0.25	40	
7	High brasses and bronzes, naval brass; Muntz metal	-0.30	45	
8	18 percent chromium type corrosion-resistant steels	-0.35	50	
9	Chromium, plated; tin, plated; 12 percent chromium type corrosion-resistant steels	-0.45	60	
10	Tin-plate; terneplate; tin-lead solder	-0.50	65	
11	Lead, solid or plated; high lead alloys	-0.55	70	
12	Aluminum, wrought alloys of the duralumin type	-0.60	75	
13	Iron, wrought, gray, or malleable; plain carbon and low alloy steels, armco iron	-0.70	85	
14	Aluminum, wrought alloys other than duralumin type; aluminum, case alloys of the silicon type	-0.75	90	
15	Aluminum, cast alloys other than silicon type; cadmium, plated and chromated	-0.80	95	
16	Hot-dip-zinc plate; galvanized steel	-1.05	120	
17	Zinc, wrought; zinc-base die-casting alloys; zinc, plated	-1.10	125	
18	Magnesium and magnesium-base alloys, cast or wrought (most anodic)	-1.60	175	

1/ Compatible couples - potential difference of 0.25 volt maximum between groups.

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6.6.3 Selection of compatible couples. Proper selection of metals in the design of equipment will result in fewer intermetallic contact problems. For example, for sheltered exposure, neither silver nor tin require protective finishes. However, since silver has an anodic index of 15 and tin 65, the EMF generated as a couple is 0.50 volt, which is not allowable by table IX. In this case, other metals or plates will be required. It should be noted that, in intermetallic couples, the member with the higher anodic index is anodic to the member with the lower anodic index and will be susceptible to corrosion in the presence of an electrolytic medium. If the surface area of the cathodic part is significantly greater than that of the anodic part, the corrosive attack on the contact area of the anodic part may be greatly intensified. Material selection for intermetallic contact parts, therefore, should establish the smaller part as the cathodic member of the couple, whenever practicable.

6.6.4 Plating. When base metals intended for intermetallic contact form couples not allowed by table IX, they are to be plated with those metals which will reduce the potential difference to that allowed by table IX.

6.6.4.1 Tin plated finishes. Use of tin plating is prohibited (see 3.4.4.1a). Use of tin finishes can result in tin whisker growth. Tin whisker growth can result in adverse effects on the operation of electronic equipment systems. For additional information on this matter, refer to ASTM B545-83 (Standard Specification for Electrodeposited Coating of Tin)."

6.7 Fixturing losses. It is known that there is inherent resistance in sockets, wiring, etc.. The qualifying activity will evaluate the manufacturer's system to determine conformance to specification requirements.

6.8 Glossary. The definitions listed below are not a complete glossary of relay terminology, but rather are intended as definitions of the technical terms as applied within the specification.

Adjunct sealant - Any hydrocarbon or silicon material used on the exterior of the unit to improve the hermeticity of a hermetically sealed relay.

Bifilar winding - Two or more windings with the wire of each winding alongside the other, matching turn for turn; may be either inductive or noninductive.

Bistable (latching) relay - A two-position relay whose contacts transfer only as a result of coil energization (depending on design) (of either a particular polarity, or of a particular coil), remain in that position with no coil energization, and transfer to the alternate position only as a result of coil energization (of either the opposite polarity, or of the other coil, respectively).

Bounce time - The interval between first make of the contact until the uncontrolled making and breaking of the contact ceases.

Chatter, contact - The undesired opening of mating contacts resulting from uncompensated ac operation, or from external shock and vibration.

Coil - One or more windings on a common form.

Contacts - The current-carrying parts of a relay that open or close electrical circuits.

Contact, bifurcated - A forked or branched, contacting member so formed or arranged, as to provide dual contacting.

Contact, closed - A normally-closed contact with the relay released is a closed contact. A normally-open contact with the relay operated is a closed contact.

Contact, open - A normally-closed contact with the relay operated is an open contact A normally-open contact with the relay released is an open contact.

Contact arrangement - The combination of contact forms that make up the entire relay switching structure.

Contact bounce - Intermittent opening of contacts after initial closure due to contact impact.

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Contact force - The force exerted by a movable contact against a mating contact when the contacts are closed.

Contact gap - The distance between a pair of mating relay contacts when the contacts are open.

Contact weld - A contact failure due to fusing of closed contacts, resulting in their failure to open.

Cycle, relay - A relay cycle is defined as a relay pickup and dropout, (one opening and one closure of a set of contacts). Therefore, a cycle consists of two operations.

Degassing - A process of thermally cleaning a relay (at elevated temperatures) to drive off harmful gasses. Degassing is usually performed using vacuum pumps.

Dropout value (voltage), specified - As the current or voltage on an operated relay is decreased, the value at or above which all relay contacts must restore to their unoperated positions.

Hermetically sealed relay - A relay contained within an enclosure that is sealed by fusion to insure a low rate of gas leakage. This usually refers to metal-to-metal, or metal-to-glass sealing.

Hold value (voltage), specified - As the current or voltage on an operated relay is decreased, the value which must be reached before any contact change occurs.

Miss - Failure to establish the intended circuit conditions (high or low output resistance, as applicable)

Normally closed contacts - Those contacts that are closed with the relay de-energized.

Normally open contacts - Those contacts that are open with the relay de-energized.

Operate time - The interval between the application of a step function input signal and closing of all normally open contacts. Bounce time is not included.

Operation, relay - A relay operation is defined as when a relay changes from an initial condition to the prescribed condition. For example, normally open to closed or vice versa. Therefore, a relay operation is one half of a cycle.

Pickup value (voltage), specified - As the current or voltage on an unoperated relay is increased, the value (voltage) at or below which all contacts must function.

Polarized relay - A relay, the operation of which is primarily dependent upon the direction (polarity) of the energizing current(s) and the resultant magnetic flux. The opposite of neutral relay. Also called "Polar Relay".

Relay - Most simply defined as an electrically controlled device that opens and closes electrical contacts to effect the operation of other devices in the same or another electrical circuit.

Relay, differential - A double input relay which is actuated by the polar signal resulting from the algebraic addition of dual inputs.

Relay, null-seeking - A relay of double-throw configuration and a stable centeroff position Either side of the double-throw contact can be activated by and for the duration of the polar actuating signal.

Relay, sensitive - A relay that operates on low input power commonly defined as 100 milliwatts or less.

Release time - The interval between the trailing edge of a step function input signal and closing of all normally closed contacts. Bounce time is not included.

Saturation - The condition attained in a magnetic material when an increase in magnetizing (coil) current produces no appreciable increase in flux.

Shelf life - A period of time a relay can be stored unused before detrimental changes occur which would cause the item not to properly perform its function within the requirements of this specification.

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Silicone - Any of a group of semi-inorganic polymers based on the structural unit R_2 SiO, where R is an organic group, characterized by wide range thermal stability, high lubricity, extreme water repellence, and physicochemical inertness, used in adhesives, lubricants, protective coatings, paints, electrical insulation, synthetic rubber, and prosthetic replacements for bodily parts.

Sensitivity - Pickup value expressed in terms of milliwatts (or watts).

Time, contact bounce - The time interval from initial closure of a contact to the end of bounce during pickup and dropout.

Time, operate - The time interval from coil energization to initial closure of the last open contact. Unless otherwise stated, it does not include contact bounce time.

Time, release - The time interval from coil de-energization to initial closure of the last open contact. Unless otherwise stated, it does not include bounce time.

Time, transfer - The time interval between the opening of the closed contact and the closing of the open contact of a break-make-contact combination. For multipole relays, transfer time is defined as the interval between the opening of the last closed contact and the closing of the first open contact of any set.

Voltage, rated coil - The coil voltage at which the relay is to operate and meet all specified electrical, mechanical and environmental requirements.

6.9 Subject term (key word) listing.

AC coil
All welded
Coil transient suppression
Full size crystal can
Half size crystal can
Hermetically sealed
Low level contact rating
Magnetic latch
Reverse polarity sensitive
TO-5

6.10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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APPENDIX A

GUIDELINES FOR CLEANING AND SMALL PARTICLE INSPECTION

10. SCOPE

10.1 Scope. The following processes, requirements, and methods of examination are offered as guidelines for performing the internal visual inspection and cleaning of relays prior to canning. These guidelines may be used both by the supplier in preparing his internal procedures and by the audit team in reviewing a supplier's procedures. This information is not intended to provide all possible requirements, processes, and test methods for internal inspections and cleaning. They are merely guidelines and should be used as such by the supplier and the audit team. Actual inspection procedures and criteria shall be in accordance with the supplier's process as approved by the qualifying activity. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS

20.1 Government documents.

20 1.1 Specifications, standards, and handbooks.

STANDARD

FEDERAL

FED-STD-209 - Clean Room and Work Station Requirements, Controlled Environment.

30. GENERAL GUIDELINES

30.1 Examination for contamination. Visually examine the following areas using the guidelines defined in detailed explanation of inspection guidelines (see 40).

a. Contact assembly, contact surfaces, stationary and movable contacts, and springs.

b. Coil, pole piece, armature, and header.

40. DETAILED EXPLANATION OF INSPECTION GUIDELINES

The suppliers procedures shall contain the following inspection points as a minimum and shall contain criteria for acceptance and/or rejection.

40.1 Moving contact assembly and springs. Inspect the moving contact assembly for proper installation and position. Any springs must clear all adjacent parts for both positions of the armature. Support brackets for the moving contact assembly shall be free of cracks and fractures (20 to 30X).

40.2 Contact surfaces (fixed and movable). Inspect surfaces for scratches or burrs in contact mating area and cracked or peeling plating.

Inspect mating contact surfaces for proper alignment for both positions of the armature. Inspect all contact areas for fibrous materials and other contaminants (20 to 30X). Inspect underside of contact supports for tool marks (20 to 30X) (see 40.6b). Inspect contact terminals for weld splatter (20 to 30X).

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40.3 Coil. Inspect coil for the following:

a. Evidence of poor weld on each coil lead wire.

b. Weld splatter at coil terminals (20 to 30X).

c. Proper lead coil dress. Ensure clearance to all moving surfaces. Coil leads should not be kinked and should not be stretched tight from coil to coil lead post (20 to 30X), i.e., adequate stress relief shall exist. Length should not be excessive.

d. Nicks in the coil lead wire due to the stripping of the insulation (20 to 30X).

e. Coil assembly for loose or frayed tape wrapping or insulation.

40.4 Armature and pole piece. Inspect armature and pole piece gap for weld splatter and contamination (20 to 30X).

40.5 Header. Inspect header (20 to 30X) for the following:

a. Unacceptable tool marks.

b. Glass seals.

c. Weld splatter.

d. Cracked or peeling plating.

e. Proper alignment of header and frame.

40.6 Inspection guidelines.

a. Weld splatter or weld expulsion balls observed under 20 to 30X magnification shall be acceptable if capable of withstanding a probing force as specified applied using an approved, calibrated force gauge. User may apply a maximum force as specified during precap inspection. Each suspect weld splatter or weld ball shall be probed only one time by the user during precap.

b. Detrimental marks, burrs, or scratches made by tooling during assembly are not acceptable.

c. Cracks in the header pin glass seals shall be deemed acceptable or unacceptable in accordance with the inspection criteria in accordance with the applicable specification.

d. Loose insulation strands from coil wrap, bobbin, and other insulators are not acceptable. Insulation strands that are an integral part and extension of the coil wrap or coil lead insulation are acceptable unless they are of sufficient length or location that they can interfere with the normal actuation and operation of the relay.

50. CLEANING

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APPENDIX A

50.1 Cleaning. Cleaning and small particle inspection shall be performed in accordance with supplier processes as approved by the qualifying activity. FED-STD-209 provides additional guidelines for limiting the amount of contamination introduced into the air in the room by closely controlling the personnel, operations, and materials inside the facility.

a. Cleaning shall be performed in a class 100 environment in accordance with FED-STD-209. Storage trays, transport trays, and covers shall be glass, non-magnetic stainless steel, or glazed ceramic. Other materials which do not attract or generate particles may be used for storage trays, transport trays and covers if approved by the qualifying activity.

b. Relays with permanent magnets shall be demagnetized, if they can be remagnetized and stabilized after canning. The relays shall be demagnetized using a suitable device.

c. Ultrasonic cleaning.

NOTE: CAUTION - Fluid used in the ultrasonic cleaner and in the small particle inspection system must be filtered finer than 1 mil to preclude contamination.

(1) Clean relay trays and covers. Clean a sufficient quantity of trays and covers for storage and transport of relays, cans, and other parts for the remainder of required cleaning. Store in class 100 environment in accordance with FED-STD-209.

(2) Ultrasonically clean relays, can and any other parts and subassemblies that constitute the final assembly. Immediately after cleaning, store the parts in covered trays in a class 100 environment in accordance with FED-STD-209.

d. Vacuum clean parts in a class 100 environment. Using a pressure gun and filtered air/gas through a static eliminator, blow filtered air/gas on the parts, holding the parts in front of a vacuum inlet to trap loosened particles. Immediately store cleaned parts in the cleaned, covered trays.

CAUTION: Nozzles used to blow filters and relays during cleaning shall not be the safety siphon type.

60. SMALL PARTICLE INSPECTION

a. Perform small particle cleaning on relays, cans and any other internal parts and subassemblies that constitute the final assembly.

(1) Obtain test liquid from prefiltered supply.

(2) Assemble a precleaned flask, vacuum pump, filter holder, precleaned 0.80 micron max filter, and precleaned funnel. Fill funnel with prefiltered test liquid and turn vacuum pump on. Repeat until flask is filled.

(3) Fill "pressurized container" with cleaned test liquid from step 60.a(2) above.

(4) Clean filter by blowing both surfaces with de-staticized air. Using the pressurized container, wash both sides of the filter with clean filtered test liquid. Observe filter under 30X magnification: If any particles .001 inch (0.03 mm) or larger are observed, repeat the cleaning process until satisfactory results are obtained. The cleaned filter shall be used in the following step 60.b.

b. Small particle inspection.

(1) Place filter holder and cleaned filter on a clean empty flask under funnel.

(2) Air blow all parts to be cleaned using de-staticized air/gas. Place parts in funnel.

(3) Using a flask of filtered test liquid, pour the test liquid into the funnel, covering the parts to be cleaned. Cover funnel. Agitation of the test liquid around the part is recommended or using a prefiltered test liquid, spray test liquid over parts into the funnel and cover funnel.

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(4) Turn vacuum pump on. When all the test liquid has passed through the filter, turn vacuum pump off.

(5) Remove filter and examine under 30X magnification.

(6) If one or more particles .001 inch (0.03 mm) or larger are present on the filter, repeat steps 60.b(1) through 60.b(5) until results are satisfactory.

70. PROCESSING OF CLEANED RELAYS

a. Place cleaned parts in cleaned covered trays in preparation for canning the relays.

b. Can relays in class 100 clean area.

NOTE: If the can or cover is subsequently removed for any reason, pre-can visual inspection shall be repeated and this cleaning procedure shall be repeated.

c. Remagnetize and stabilize the relays if applicable.

d. If relays are required to be removed from the class 100 clean area, place a gummed clean label over the evacuation hole (if applicable) before removing from clean area.

e. Relays shall be manufactured, assembled, and tested in controlled areas. After final cleaning, all processes and assembly performed on the relay while in a delidded configuration shall be accomplished on a class 100 laminar flow bench or in an area meeting the requirements of a class 100 clean room as defined in FED-STD-209 If a final cleaned relay leaves the flow bench or clean room, relay shall be lidded and evacuation hole(s) shall be covered by a method approved by the qualifying activity. Additional provisions shall be incorporated as required in all areas to prevent the generation and entry of particulate contamination into parts and assemblies. Additional provisions shall also be made as required in all areas to remove particulate contamination from the parts and assemblies. The supplier shall maintain records which verify that the required levels of cleanliness are maintained in all assembly and sealing areas.

80. REWORK

80.1 Rework. Relays that require rework or readjustment after capping, but prior to any welding of the enclosure, may be reworked or readjusted provided the examination and cleaning of 40 through 70 are repeated after the rework or readjustment.

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APPENDIX B

PARTICLE IMPACT NOISE DETECTION (PIND)

10. SCOPE

10.1 Scope. The purpose of this test is to detect the presence of free moving particulate contaminants within sealed cavity devices. This test method is specifically directed toward relays and other devices where internal mechanism noise makes rejection exclusively by threshold level impractical. The test provides a nondestructive means of identifying those devices containing particles of sufficient mass that, upon impact with the case, excite the transducer. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS This section is not applicable to this appendix.

30. EQUIPMENT

30.1 Equipment. The equipment required for the particle impact noise detection (PIND) test shall consist of the following (or equivalent). PIND instruments are available which incorporate items a - h (see figure 11).

a An oscilloscope capable of 500 kHz response minimum, and a sensitivity of 20 mV/cm for visual display of the particle noise.

b An audio system with speaker to monitor the audio signal from the PIND electronics. If headphones are used, the system shall provide safeguards against loud noise bursts.

c. A vibration shaker and driver assembly to accommodate the weight of the payload. The payload consists of the DUT (device under test), (PIND) transducer, the transducer isolator, preamplifier (when included), co-test shock mechanism (when included), a portion of the transducer cable and its restraints. The shaker and driver assembly shall be capable of providing essentially sinusoidal motion at.

(1) Condition A. 5g peak at 27 Hz in 3 axes.

(2) Condition B: 5g peak at 27, 40, and 100 Hz in optimum axis.

d. PIND transducer, calibrated to a peak sensitivity of -77.5 ±3 dB at 1 volt per microbar at a point within the frequency range of 150 - 160 kHz.

e. A sensitivity test unit (STU) (see figure 11) for periodic assessment of the PIND system performance. The STU shall consist of a transducer with the same tolerances as the PIND transducer and a circuit to excite the transducer with a 250-microvolt ±20 percent pulse. The STU shall produce a pulse of about 20 mV peak on the oscilloscope when the transducer is coupled to the PIND transducer with attachment medium.

f. PIND electronics, consisting of an amplifier with a gain of +60 ±2 dB centered at the frequency of peak sensitivity of the PIND transducer to amplify the transducer signal to a usable level for audio detection and oscilloscope display. The noise at the output of the amplifier shall not exceed 10 mV peak

g. Co-test shock mechanism consisting of the integral co-test shock mechanism of 30.1c above (when included) capable of imparting shock pulses of at least 200g peak to the DUT, and shall be capable of providing a pulse duration not exceeding 1 ms. If the integral co-test shock system is used, the shaker vibration may be interrupted or perturbed for a period of time not to exceed 250 ms from initiation of the shock pulse.

NOTE: When co-test shock capabilities are not available, the DUT shall be pre-shocked in accordance with 40 4.1b

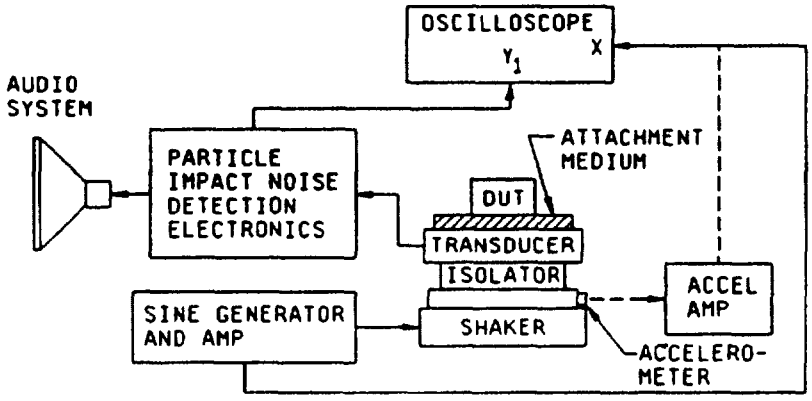
h Isolator material between the PIND transducer and the vibration shaker and driver when required to reduce background noise. The isolator shall have no resonance within the test frequency range.

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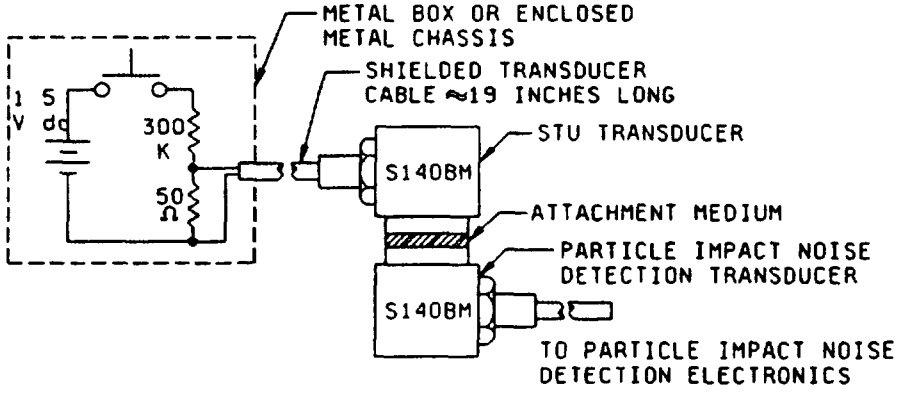
APPENDIX B

i. Attachment medium. The attachment medium used to attach the DUT to the PIND transducer shall be either a viscous acoustic couplant or double-faced tape. A mechanical holding fixture that can be shown to produce no noise signature may also be used, in conjunction with acoustical couplant (if necessary) for large packages.

j. Special mounting adapters for devices which have irregular surfaces (see 40.4.1)



Typical particle impact noise detection system



Typical sensitivity test unit

NOTES:
1. Pushbutton switch: Mechanically quiet, fast make, gold contacts. E.G. T2 SM4 microswitch.
2. Resistance tolerance 5 percent noninductive.
3. Voltage source can be a standard dry cell.
4. The coupled transducers must be coaxial during test.
5. Voltage output to STU transducer 250 microvolts, ±20 percent
6. A function generator with a 250 microvolt ±20 percent square wave output at 150 kHz nominal with 50 ohms output impedance may be substituted for enclosed metal chassis.

FIGURE 11. Typical particle impact noise detection system and sensitivity test unit.

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APPENDIX B

40. PROCEDURE

40.1 Test equipment setup. The test equipment shall be set up in a low noise area. Noise in this context refers not only to that audible noise which would normally interfere with listening but also power line noise, radiated R.F. noise or high frequency acoustic noise (150 - 160 kHz). Commercial equipment shall be connected as described in the operations manual. Otherwise, assembled PIND equipment shall be connected as shown on figure 11.

a. Audio output volume shall be adjusted to a comfortable noise level output.

b. Shaker drive frequency shall be adjusted in accordance with 40.3.

c. Shaker drive amplitude shall be adjusted in accordance with 40.3 and mounting adapter (if any) shall be in place.

d. Oscilloscope vertical deflection primary beam sensitivity (displaying PIND electronics output) shall be 20 millivolts/centimeter Oscilloscope horizontal deflection shall be adjusted to 4 cm and shall obtain drive from the sine generator/amplifier accelerometer, or a time base (2 ms/cm) triggered from the accelerometer output.

40.2 Test equipment checkout. The test equipment checkout shall be performed daily or prior to the start of, and at the completion of, daily PIND testing with results recorded. Failure of the system to meet checkout requirements shall require retest of all devices tested subsequent to the last successful system checkout.

40.2.1 Shaker drive system checkout. The drive system shall achieve the shaker frequency and the shaker amplitude specified in 40.3. The drive system shall be calibrated so that the frequency settings are within ±8 percent and the amplitude vibration settings are within ±10 percent of the nominal values. If a visual displacement monitor is affixed to the transducer, it may be used for amplitudes between .04 and .12 inch (1.02 and 3 05 mm). An accelerometer may be used over the entire range of amplitudes and shall be used below amplitudes of 040 inch (1.02 mm).

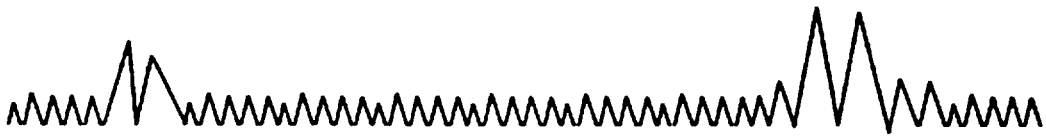
40.2.2 Detection system checkout. With the shaker de-energized, the STU transducer shall be mounted face-to-face and coaxial with the PIND transducer using the recommended attachment medium. The STU shall be activated several times to verify low level signal pulses visually on the oscilloscope (approximately 20 mV peak or 10 mV peak above system noise). If a commercial instrument is used, follow the manufacturer's detection system checkout procedure. With the approval of the qualifying activity, an alternate calibrated noise generator may be used to perform the system checkout. In this case, the calibrated generator is tested in accordance with this procedure to demonstrate that oscilloscope traces are achieved to result in rejection.

NOTE: Not every application of the STU will produce the required amplitude but the majority of applications will do so.

40.2.3 System noise verification. For proper system operation, no extraneous noise can be permitted to exist in the system. During proper operation, the normal system noise, as observed on the oscilloscope, will appear as a fairly constant band and must not exceed 10 mV zero to peak. Extraneous noise is defined as noise in the system other than the permissible background noise that is present with no device on the transducer. Such noise can be due to a number of sources which must be eliminated or their effects guarded against, since those non-signal noise spikes can appear as signals on the indicators. Common sources of noise are fluorescent lighting, heater elements, soldering irons and other switching transients, line transients and, especially, less than optimum installation and support of the transducer cabling. The latter source normally may be eliminated by redressing the cable, tightening or cleaning the connector at the transducer, or even replacing the transducer or transducer cable. To verify that no extraneous noise exists in the system, observe the oscilloscope while turning on the shaker at the designated frequency and amplitude. This noise is usually present as pulses which remain in a fixed position on the oscilloscope trace If extraneous noise is observed, correct the problem by shielding or other precautions, such as those suggested above and re-run the entire noise check.

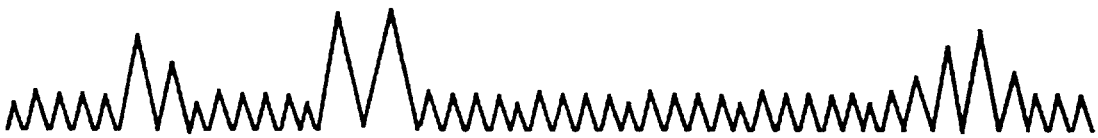
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Acceptance criteria: Each unit tested shall meet the acceptance.



Shaker noise: Timebase adjusted to locate shaker reversal noise bursts at end of oscilloscope trace. Test unit not mounted.

a



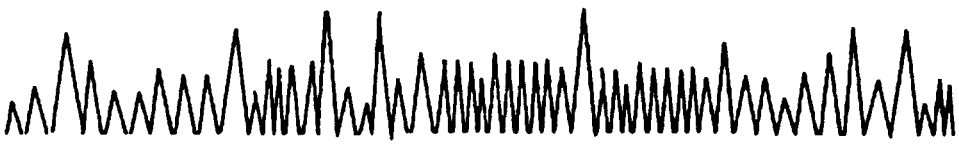
Inherent mechanical noise Synchronized spike may appear at different locations on timebase for each unit under test.

b



Particulate noise: Nonsynchronized spikes of any magnitude appear randomly and may disappear as test progresses Unit is rejectable

c



Excessive mechanical noise:

Synchronized trace masks more than 50 percent of oscilloscope trace. Unit is rejectable.

d

FIGURE 12. Representative oscilloscope traces

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40.3 Test parameters. The following test parameters shall be applied to each DUT. If it has been demonstrated that one or more of these parameters are damaging to a given part type, the test shall be performed as defined by the parts detailed specification. Test parameters shall be documented in the report.

40.3.1 Vibration frequency:

Condition A: 27 Hz, three axes.

Condition B: 27 Hz, 40 Hz, and 100 Hz, optimum axis.

40.3.2 Vibration amplitude. Conditions A and B: 5g.

40.3.3 Shock level. Conditions A and B: 200g peak, 1 ms, maximum.

40.4 Test sequence. The condition A or B (as applicable) test sequence shall be applied to each DUT as follows:

40.4.1 Condition A sequence.

a. Vibration 5 seconds at 27 Hz.

b. Co-test shock (three pulses).

NOTE: When co-test shock capability does not exist, the DUT shall be preshocked with 2 pulses in each of 3 axes at 200g peak, 1 ms (maximum), using equipment and techniques as defined by method 213 of MIL-STD-202, and step 40 4.1b. may be omitted.

c. Vibration 5 seconds at 27 Hz.

d. Repeat test in the two other orthogonal axes.

e. Accept or reject.

40.4.2 Condition B sequence. Apply 40.4 1, steps a, b, and c to the DUT in its optimum axis as defined by the detailed specification. Repeat the test at 40 Hz and 100 Hz. Accept or reject.

40.5 Mounting requirements. For condition A, the DUT shall be mounted such that it is ultimately tested in three orthogonal axes, one of which represents the normal mounting configuration for the device For condition B, the DUT shall be mounted in the optimum axis as specified by the detail specification for minimizing inherent device noise. When special fixturing is required for unusual package configurations, such fixtures shall have the following properties.

a. Low mass.

b. High acoustic transmission (aluminum alloy 7075 works well).

c. Full transducer surface contact, especially at the center.

d. Maximum practical surface contact with test part.

e. No moving parts.

f. Suitable for attachment medium mounting.

Leads on the parts shall be dressed, as necessary, so they will not strike each other or the transducer during vibration. Long or thin cross section leads shall be observed for signs of resonance, indicated by motion exceeding 3 or 4 diameters. Such resonance may give extraneous noise during test even though the leads do not strike each other In these cases, the leads may have to be shortened (if permitted by the application) or special fixturing or frequency changes may be required

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40.6 Test monitoring. To avoid false indications, the DUT shall be inspected for any attached foreign matter or leads which are touching each other. The DUT shall be mounted on the center of the transducer using attachment medium or, if necessary, a mounting adapter or holding fixture. To provide maximum signal transmissibility with a viscous couplant, a sufficient amount of couplant shall be used and the DUT shall be firmly mounted so that any excess couplant can be squeezed out. When double-faced tape is used, it shall be changed at the start of a test group and after 25 units or less thereafter. Devices shall be put on and removed from the attachment medium with a slight twisting motion. Device orientation for each package type shall be as specified in 40.5. The vibration frequency and the vibration amplitude shall be set at the level specified in 40.3. Both detection systems shall be monitored for evidence of loose particles. Any device which gives a particle indication shall be considered a reject. Particle indications can occur in either detection system as follows:

a. Visual indication of high frequency spikes which exceed the normal constant background white noise level and the DUT inherent noise level.

NOTE: The repeatable periodic inherent noise signature of electromechanical devices must be fully characterized to avoid false indications of failure and to avoid the masking of particles (see figure 11).

b Audio indication of clicks, pops, or rattling which is different from the noise signature of the DUT or the constant background noise present with no DUT on the transducer.

c If there is no indication of particles prior to co-test shock or 5 seconds after co-test shock, the device is acceptable. When pre-shock is used in place of co-test shock, there shall be no particle indications during the 10 seconds of vibration

40.7 Co-test shock application (when applicable) The operation of the co-test shock mechanism shall be in accordance with procedures supplied by the equipment manufacturer. In systems that disable the detector during the co-test shock, the period of time from shock pulse to reinitiation of detection shall not exceed 100 ms.

40.8 Additional test cycles. If additional cycles of testing on a lot are specified, the entire test procedure (equipment set-up and checkout mounting, vibration, and co-shocking) shall be repeated for each retest cycle. Reject devices from each test cycle shall be removed from the lot and shall not be retested in subsequent lot testing.

40 9 Failure criteria. Any noise bursts as detected by either of the two detection systems (see 40.6) exclusive of DUT inherent noise or background noise during the monitoring periods shall be cause for rejection of the device. Rejects shall not be retested, except for retest of all devices in the event of test system failure as provided for 40.2.

50. SUMMARY

50.1 Summary. The following details shall be specified in the applicable detail specification or procurement documentation.

a. Lot acceptance/rejection criteria (if applicable)

b Vibration, conditions A or B

NOTE Unless otherwise specified, condition A shall apply When condition B is required, the optimum (required) axis of vibration shall be specified

c Co-test shock level and duration, if other than as defined herein.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR, MI
Navy - AS, MC, OS
Air Force - 17, 19, 99

Preparing activity:
DLA - ES

(Project 5945-0878-01)

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-R-39016E	2. DOCUMENT DATE (YYMMDD)

3. DOCUMENT TITLE

RELAYS, ELCTROMAGNETIC, ESTABLISHED RELIABILITY, GENERAL SPECIFICATION FOR

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Commander Defense Electronics Supply Center ATTN: DESC-ELD	b TELEPHONE (Include Area Code) (1) Commercial 513-296-6184 (2) AUTOVON 986-6184
c. ADDRESS (Include Zip Code) 1507 Wilmington Pike Dayton, OH 45444-5765	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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